Social Network Analysis (**SNA**)– Fall 2025

Uncovering Structural Patterns in U.S. Air Travel: A Network-Based Study of Passenger Movements Across States

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Abstract:

This study applies Social Network Analysis (SNA) to examine the structural dynamics of the U.S. domestic airline network from 2019 to 2022, capturing the periods before, during, and after the COVID-19 pandemic. By modeling U.S. states as nodes and passenger flows as weighted, directed edges, the research investigates network centrality, bottlenecks, influence, regional connectivity, and passenger flow dependency. Using key SNA metrics—including degree centrality, weighted betweenness, closeness, eigenvector centrality, clustering coefficients, assortative, and passenger flow dependency and diversity indices—the study identifies persistent hub states and evaluates how network structure responded to systemic shock and recovery.

The results reveal a highly centralized, disassortative hub-and-spoke system dominated by a small group of metropolitan states, particularly California, Texas, Florida, New York, and Illinois. While the pandemic caused a sharp contraction in passenger volumes and temporarily altered relative dominance among hubs, the underlying network structure proved remarkably resilient. Regional connectivity became more prominent during the crisis, highlighting the stabilizing role of localized airline networks.

Overall, the findings demonstrate that COVID-19 disrupted traffic intensity more than structural hierarchy, offering important insights into airline network resilience, vulnerability, and recovery dynamics.

Introduction:

Air transportation in the United States is one of the most complex and interconnected mobility systems in the world, with millions of passengers traveling between states each year. Understanding how states interact through air travel is essential for identifying major hubs, critical corridors, structural vulnerabilities, and patterns of passenger flow, especially following major disruptions such as the COVID-19 pandemic.

Social Network Analysis (SNA) provides a powerful framework to study these interactions by representing states as nodes and passenger flows as weighted connections. Using this approach allows for the identification of central states, bottlenecks, influential hubs, and structural patterns that impact the efficiency and resilience of the network. This project applies SNA to the U.S. domestic airline network to examine connectivity, detect key hubs and routes, and analyze passenger flow dependency and diversity, providing insights that support both network planning and resilience assessment.

Background:

Theoretical background:

Airlines often follow a **hub-and-spoke model**, where a few major states act as central hubs that gather and distribute to passengers. This creates efficiency but also makes the network vulnerable: disruptions at major hubs can affect many connected states.

Passenger flows depend on population size, economic activity, distance, and accessibility. Large economic states (e.g., California, Texas, Florida) naturally become major travel attractors.

Air transport is also tied to **regional development theory**, as states with stronger air connectivity tend to enjoy greater economic activity, tourism, and mobility. This creates geographic inequalities, with some regions much better connected than others.

Transportation resilience theory highlights that airline networks are **efficient but fragile**. Events like COVID-19 can significantly alter connectivity, centralization, and passenger volumes, making it important to analyze changes across years.

Overall, these theories explain why certain states become hubs, why passenger flows vary across routes, and how external shocks can reshape the structure of the U.S. airline network. They provide the foundation for analyzing centrality, connectivity, bottlenecks, and structural changes in your research.

Research Questions and Motivation:

1) Which states are most central in the US airline network?

2) Which states are bottlenecks that most traffic must pass through?

3) Which states are most influential in spreading disruptions or delays?

4) Which states serve as "air travel influencers" whose connectivity boosts nearby states?

5) Which states maximize short-range regional connectivity?

6) Which states serves as the dominant origin and destination in the US airline network, exhibiting the highest total passengers landing and total passengers departing? And did it change after the pandemic and during the recovery years or not?

7) Do major hubs in the U.S. airline network tend to connect mostly with other hubs or with smaller states?

8) Passenger Flow Dependency and Diversity

Scope of the study:

This project analyzes:

- **U.S. domestic air passenger flows** between 2019–2022
- **State-to-state connections** (not individual airports)
- Metrics calculated using **Gephi and Network x.**
- Focus on network structure, centrality, resilience, and regional influences

The scope centers strictly on **state-level passenger network behavior**

Project Goals:

The primary goal of this project is to use Social Network Analysis to examine the structure, connectivity, and dynamics of the U.S. airline network. The study aims to identify the most central states that serve as key hubs facilitating the highest levels of passenger connectivity and to detect bottleneck states where most passenger traffic passes, highlighting critical points that could affect network efficiency.

It also seeks to determine the states most influential in spreading disruptions or delays, as well as "air travel influencers," whose connectivity enhances the accessibility of neighboring states and regional mobility.

Furthermore, the project analyzes hub connectivity patterns to assess whether major hubs tend to connect primarily with other hubs or with smaller, less-connected states, revealing the overall network structure.

Finally, it measures passenger flow dependency and diversity to understand the concentration of traffic and the resilience of each state's outbound connections.

Methodology:

The data were aggregated at the state level instead of the city level to reduce network complexity and focus on broader connectivity patterns. This approach allows for clearer interpretation of national and regional passenger flows and provides more stable and comparable results across time. It also helps identify key hub states, bottlenecks, and structural patterns in the U.S. airline network.

Social Network Analysis (SNA) is ideal for this study because it models states as nodes and passenger flows as weighted edges, capturing the networked nature of U.S. air travel. It allows the identification of central hubs, bottlenecks, influential states, and weak connections, while also measuring regional clustering, connectivity, and resilience.

SNA provides both quantitative metrics and visual tools to analyze structural patterns, changes over time, and the effects of disruptions—insights that traditional statistical or geographic methods cannot fully reveal.

Additionally, aggregating data at the state level results in the presence of self-loops, which represent passenger movements occurring within the same state. These self-loops capture intra-state air travel activity and were retained to preserve the completeness of passenger flow patterns and overall network structure.

Data Collection:

Step 1: Data Collection

- Dataset: US Airline Flight Routes and Fares (1993–2024)
- Filtered to **2019–2022 only**
- Selected variables: Origin State, Destination State, Passenger Count

Step 2: Data Preparation

- Aggregated passenger totals for each state-to-state pair
- Converted into an edge list:
    - **Source state**
    - **Target state**
    - **Passenger weight**
- Created node list:
    - **Id**
    - **State name (abbreviation)**

Step 3: Graph Construction

- Nodes = states
- Edges = state-to-state weighted connections
- Graph type = directed Weighted Network

The States Abbreviation:

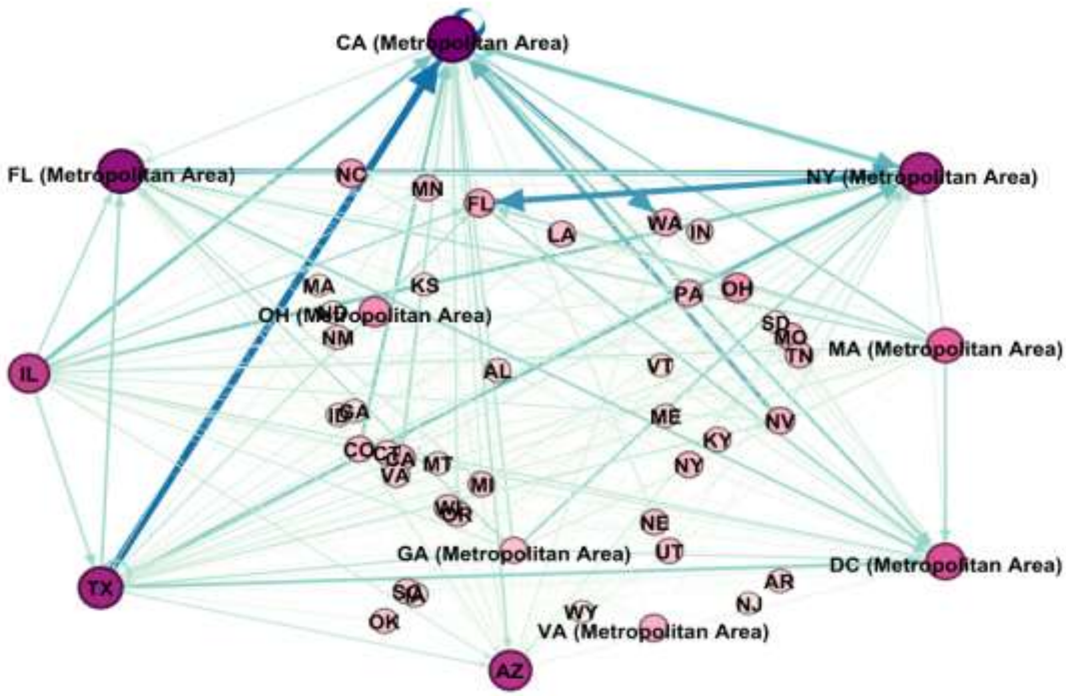
State abbreviation	State name
AL	Alabama
AR	Arkansas
AZ	Arizona
CA	California
CA (Metropolitan Area)	California (Metropolitan Area)
CO	Colorado
CT	Connecticut
DC (Metropolitan Area)	District of Columbia (Metropolitan Area)
FL	Florida
FL (Metropolitan Area)	Florida (Metropolitan Area)
GA	Georgia
GA (Metropolitan Area)	Georgia (Metropolitan Area)
IA	Iowa
ID	Idaho
IL	Illinois
IN	Indiana
KS	Kansas
KY	Kentucky
LA	Louisiana
MA	Massachusetts
MA (Metropolitan Area)	Massachusetts (Metropolitan Area)
ME	Maine
MI	Michigan
MN	Minnesota
MO	Missouri
MS	Mississippi
MT	Montana
NC	North Carolina

ND	North Dakota
NE	Nebraska
NJ	New Jersey
NM	New Mexico
NV	Nevada
NY	New York
NY (Metropolitan Area)	New York (Metropolitan Area)
OH	Ohio
OH (Metropolitan Area)	Ohio (Metropolitan Area)
OK	Oklahoma
OR	Oregon
PA	Pennsylvania
SC	South Carolina
SD	South Dakota
TN	Tennessee
TX	Texas
UT	Utah
VA	Virginia
VA (Metropolitan Area)	Virginia (Metropolitan Area)
VT	Vermont
WA	Washington
WI	Wisconsin
WY	Wyoming

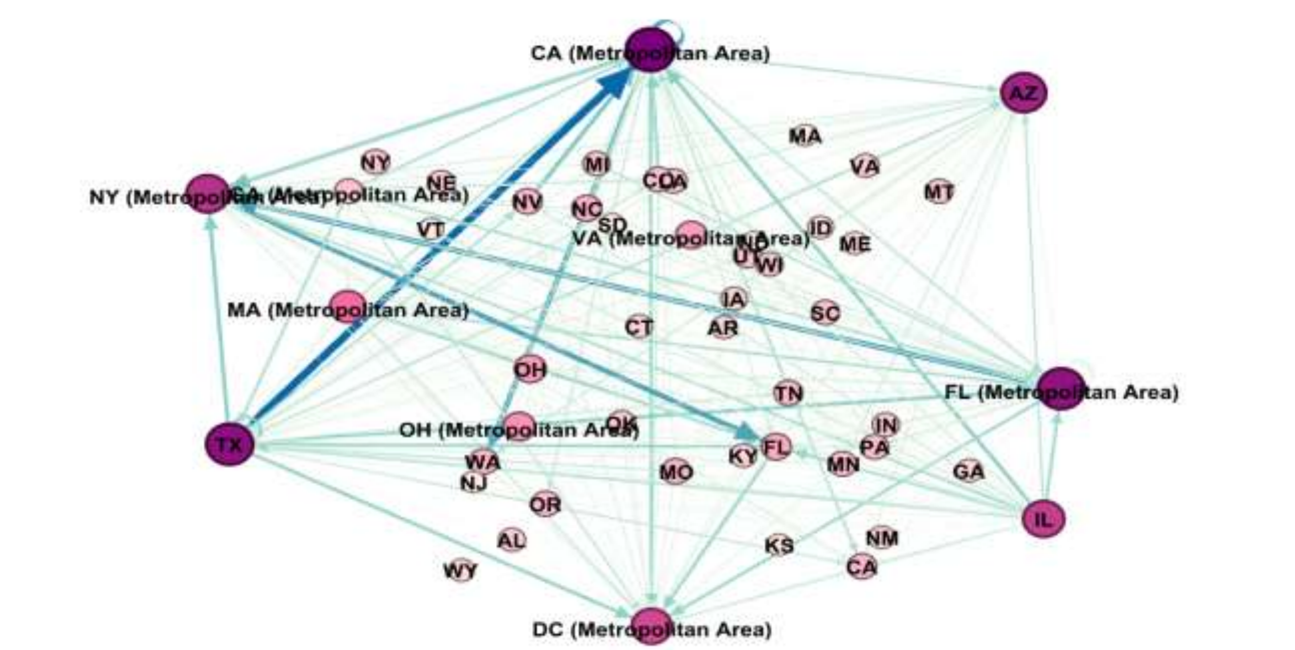
Research Questions:

1) **Which states act as the most important hub for passenger traffic?**

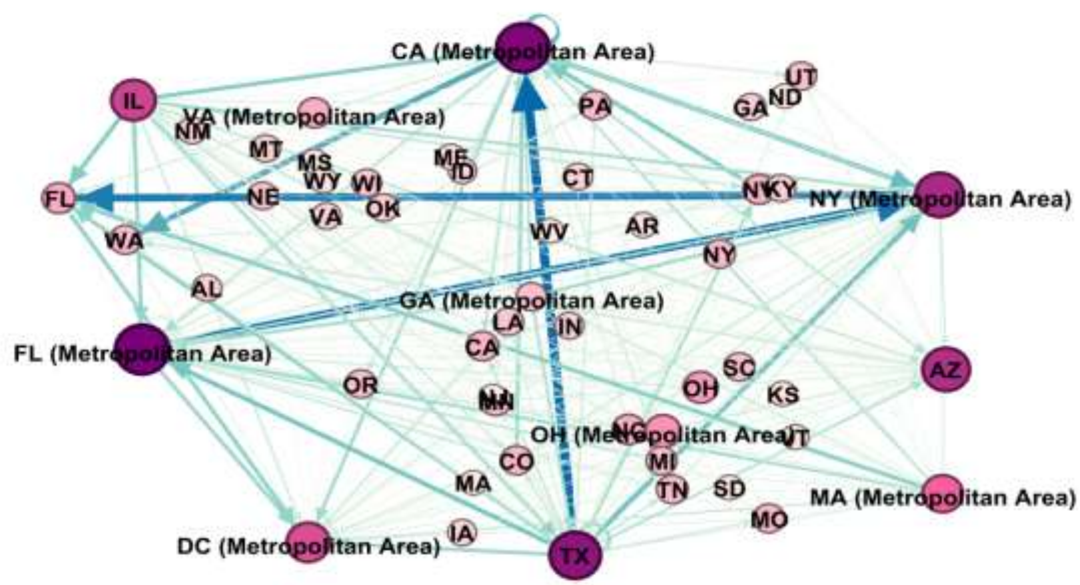
Pre-COVID-19 (Year 2019)



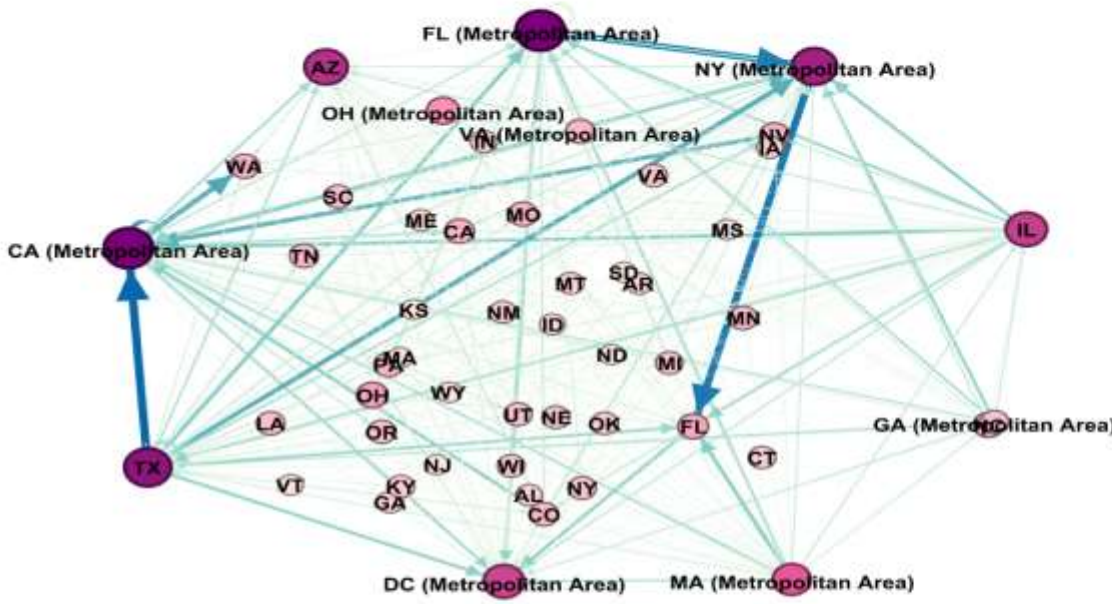
During COVID-19 (Year 2020)



Early Recovery (Year 2021)



Post-COVID-19 (Year 2022)



Node's color from low to high based on node's degree

Edge's color from low to high weighted by number of passengers

Comparing states' degree and the weighted degree

(where weights represent the number of passengers)

2019			2020		
Id	Degree	Weighted Degree	State	Degree	Weighted Degree
CA (Metropolitan Area)	65	1,086,062	CA (Metropolitan Area)	67	344,954
NY (Metropolitan Area)	53	707,834	TX	62	253,439
TX	54	638,200	NY (Metropolitan Area)	51	200,213
FL (Metropolitan Area)	59	461,611	FL (Metropolitan Area)	61	197,725
IL	46	427,126	IL	47	133,517
DC (Metropolitan Area)	40	402,666	DC (Metropolitan Area)	44	126,639
MA (Metropolitan Area)	35	313,749	AZ	56	116,725
FL	17	258,986	FL	17	101,683
AZ	49	248,299	MA (Metropolitan Area)	33	88,785
GA (Metropolitan Area)	11	168,235	GA (Metropolitan Area)	11	64,172

2021			2022		
Id	Degree	Weighted Degree	Id	Degree	Weighted Degree
CA (Metropolitan Area)	64	607,404	CA (Metropolitan Area)	63	881,713
TX	60	463,946	NY (Metropolitan Area)	53	655,448
NY (Metropolitan Area)	53	426,566	TX	57	623,009
FL (Metropolitan Area)	66	392,641	FL (Metropolitan Area)	63	500,821
IL	44	275,453	IL	44	379,085
DC (Metropolitan Area)	41	227,582	DC (Metropolitan Area)	42	341,482
FL	17	215,787	FL	16	282,507
AZ	51	197,186	MA (Metropolitan Area)	36	267,322
MA (Metropolitan Area)	35	173,739	AZ	48	254,035
GA (Metropolitan Area)	11	117,943	NV	14	153,632

Interpretation:

Based on passenger-weighted degree centrality (California (Metropolitan Area), Texas, New York (Metropolitan Area), and Florida (Metropolitan Area) consistently act as the most important hubs for passenger traffic in the US airline network.

The accompanying table shows that these states record the highest weighted degree values across the four-year period, indicating that they handle the largest volumes of passenger flows.

The network visualization for the four years (2019,2020,2021 and 2022) further illustrates a sharp decline in passenger traffic in 2020, reflecting the impact of the COVID-19 pandemic and the associated travel restrictions. Despite this temporary contraction, the same major hubs remain centrally positioned within the network.

The observed recovery in passenger-weighted centrality in the later years, as reflected in the visual comparison of the graphs, suggests that passenger traffic gradually rebounded, with structurally important hubs experiencing faster recovery than less connected states.

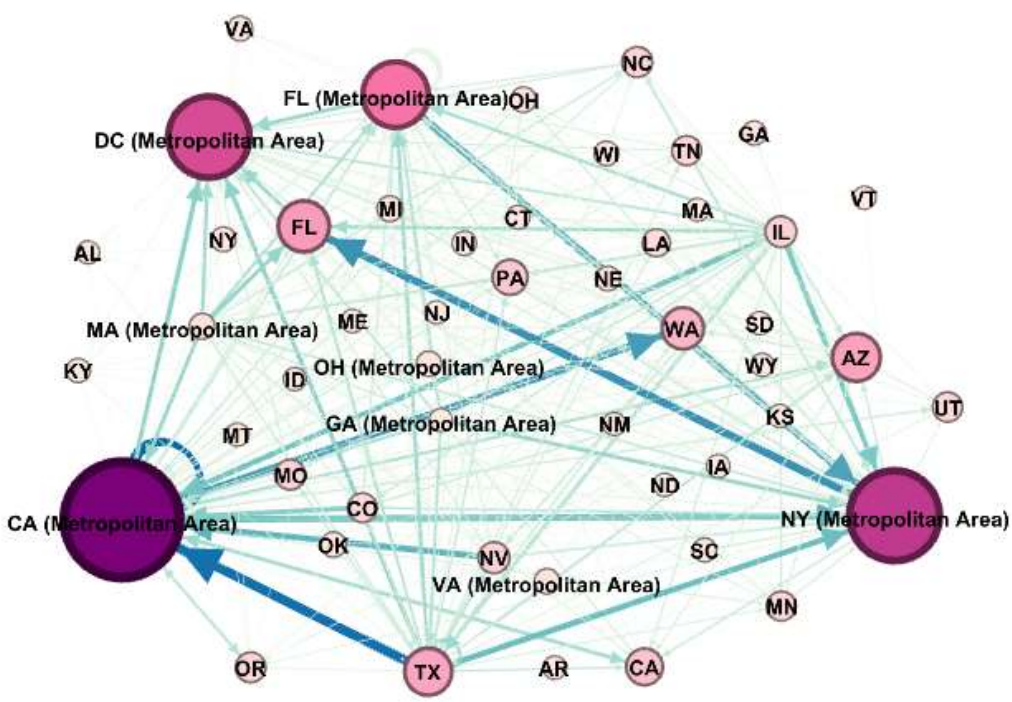
Overall, the table and the network visualization jointly confirm the dominant role of these states in shaping passenger flows over time.

2) Which states serve as the dominant origin and destination in the US airline network, exhibiting the highest total passengers landing and total passengers departing? And did it change after the pandemic and during the recovery years or not?

During COVID-19 (Year 2020)

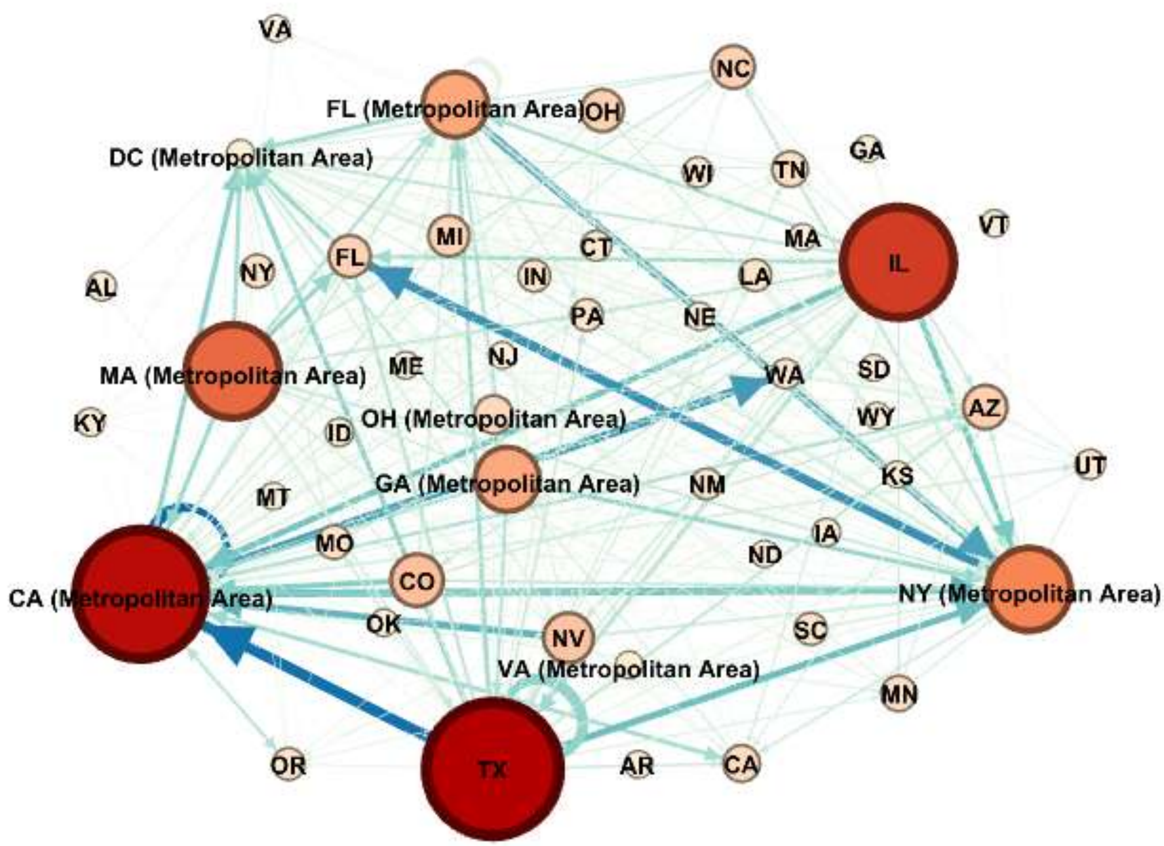
1) passengers landing

Id	Weighted In-Degree
CA (Metropolitan Area)	634061.0
NY (Metropolitan Area)	458360.0
DC (Metropolitan Area)	398696.0
FL (Metropolitan Area)	291098.0
FL	191838.0
AZ	171044.0
TX	165347.0
WA	127985.0
CA	93806.0



2) passengers departing

Id	Weighted Out-Degree
TX	472853.0
CA (Metropolitan Area)	452001.0
IL	371944.0
MA (Metropolitan Area)	295891.0
NY (Metropolitan Area)	249474.0
FL (Metropolitan Area)	170513.0
GA (Metropolitan Area)	168235.0
CO	114103.0
NV	102928.0



Interpretation:

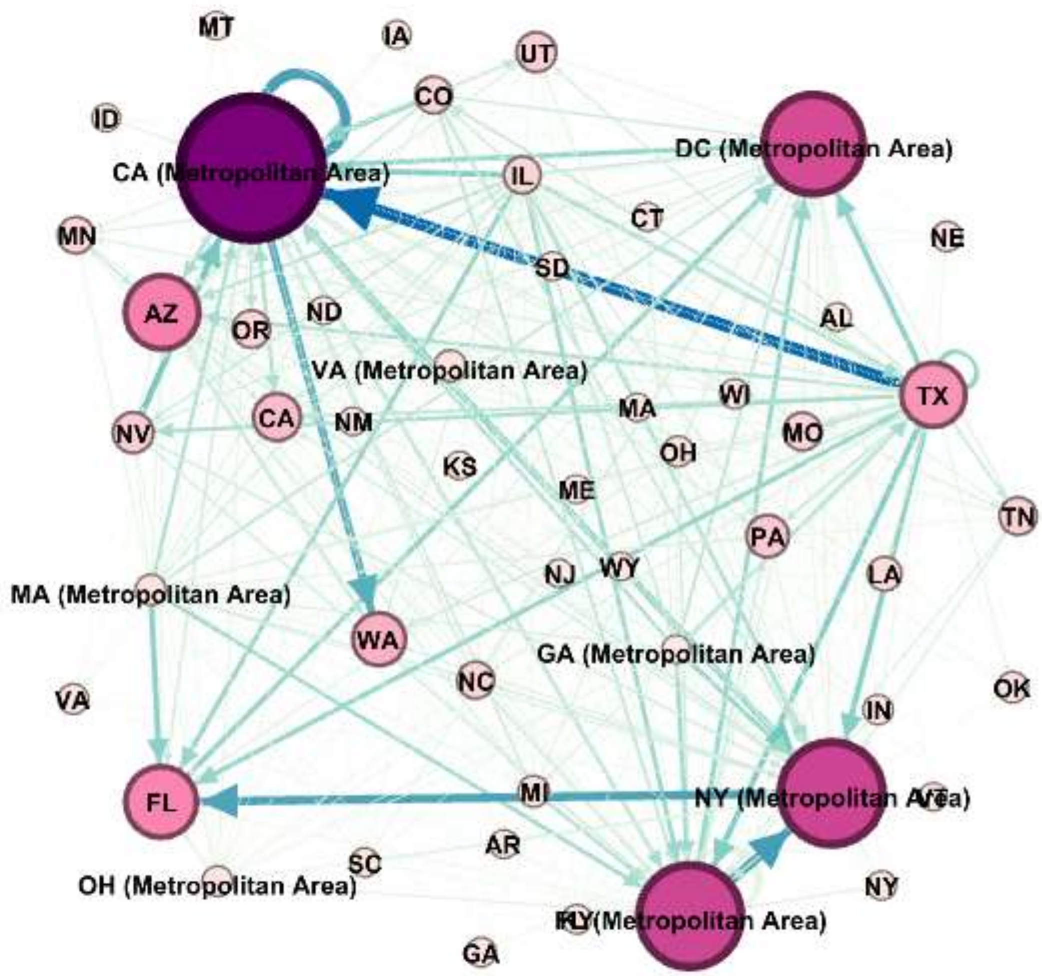
In the pre-pandemic baseline, the US airline network functioned as a highly optimized, high-volume system defined by structural symmetry. The states of **California (CA), Texas (TX), Florida (FL), New York (NY), and Illinois (IL)** served as the undisputed "Super-Nodes" for both passengers landing (Weighted In-Degree) and departing (Weighted Out-Degree). The visualization for this year reflects a dense, cohesive mesh where traffic was evenly distributed across these geographically distinct hubs, indicating a balance between business travel **(NY, IL)** and leisure/gateway traffic **(FL, CA)**.

Critically, the network in 2019 exhibited near-perfect equilibrium between arrivals and departures for these top states. The "passengers landing" map and "passengers departing" map are virtually identical, confirming that the network operated efficiently with minimal dead-heading or asymmetric flow. This stability established a hierarchy where the dominant states acted as massive dual-purpose transfer points, funneling traffic from smaller regional spokes to the rest of the country and the world.

During COVID-19 (Year 2020)

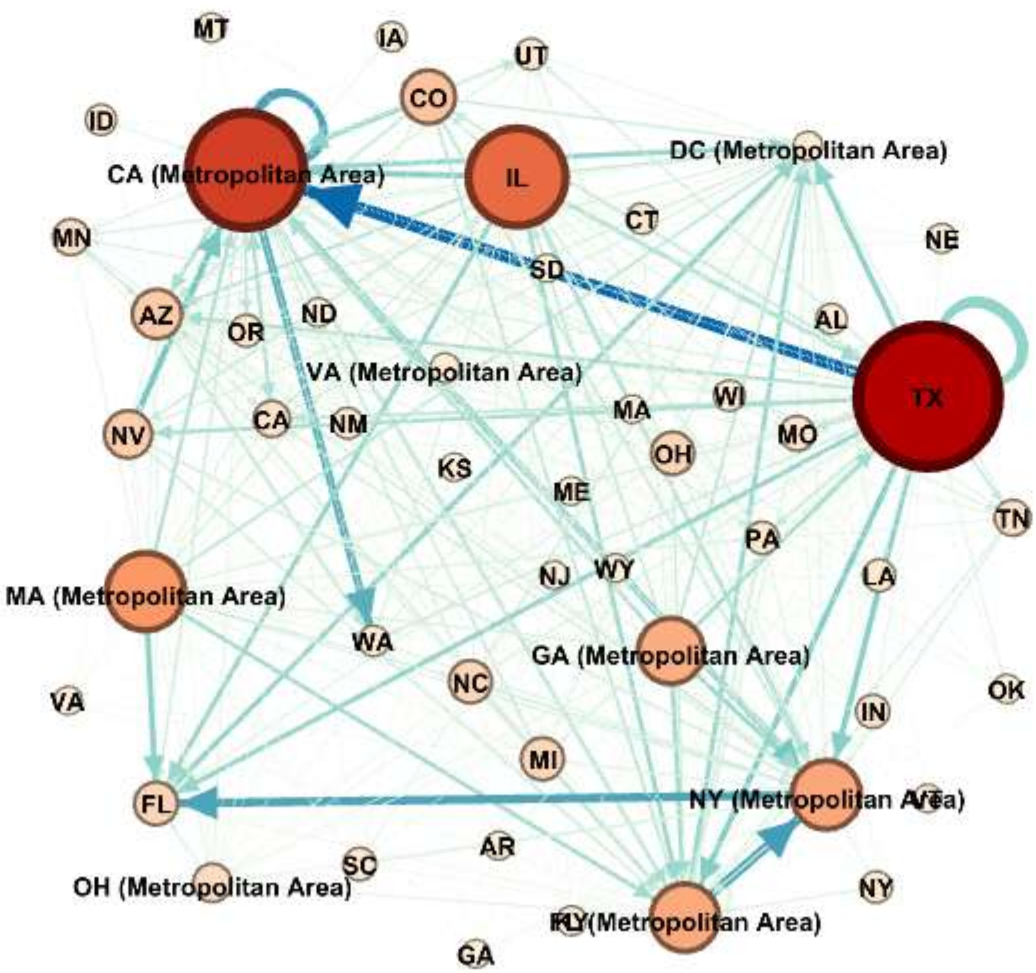
1) passengers landing

Id	Weighted In-Degree
CA (Metropolitan Area)	197148.0
NY (Metropolitan Area)	130793.0
FL (Metropolitan Area)	130135.0
DC (Metropolitan Area)	124699.0
AZ	80338.0
FL	75753.0
TX	63302.0
WA	43558.0
CA	33953.0



2) passengers departing

Id	Weighted Out-Degree
TX	190137.0
AR	895.0
CA	14095.0
IN	9629.0
KS	0.0
MO	11690.0
NC	25053.0
NV	33177.0
OK	1535.0



Interpretation:

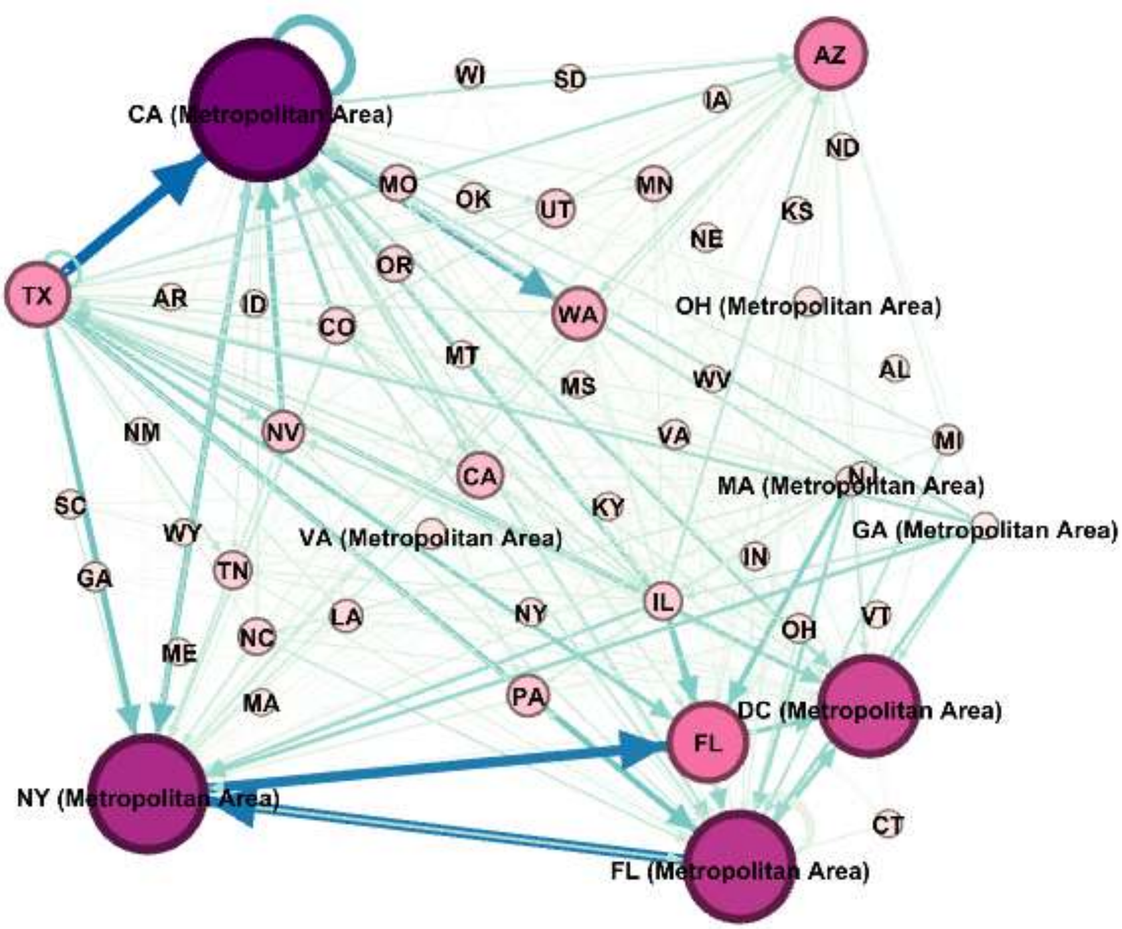
The 2020 network reveals a dramatic "hollowing out" of the system, where the absolute volume of passengers collapsed, and the relative hierarchy of dominant states shifted significantly. While the physical infrastructure remained, the dominance of **New York (NY)** and **Massachusetts (MA)**—key business and international gateways—waned considerably due to strict lockdowns and the cessation of corporate travel. In their place, **Florida (FL)** and **Texas (TX)** emerged as the most resilient nodes, maintaining higher relative Weighted In-Degree and Out-Degree because they remained open and continued to serve domestic leisure travelers when other regions froze.

Structurally, this year shows the highest potential for asymmetry and fragmentation. The "passengers landing" vs. "passengers departing" maps likely show subtle discrepancies as airlines scrambled to reposition aircraft and adjust to erratic demand. The network temporarily devolved from a cohesive national mesh into a more fractured set of regional sub-graphs, heavily reliant on the few remaining active hubs in the South and Southwest to maintain essential connectivity.

Early Recovery (Year 2021)

1) passengers landing

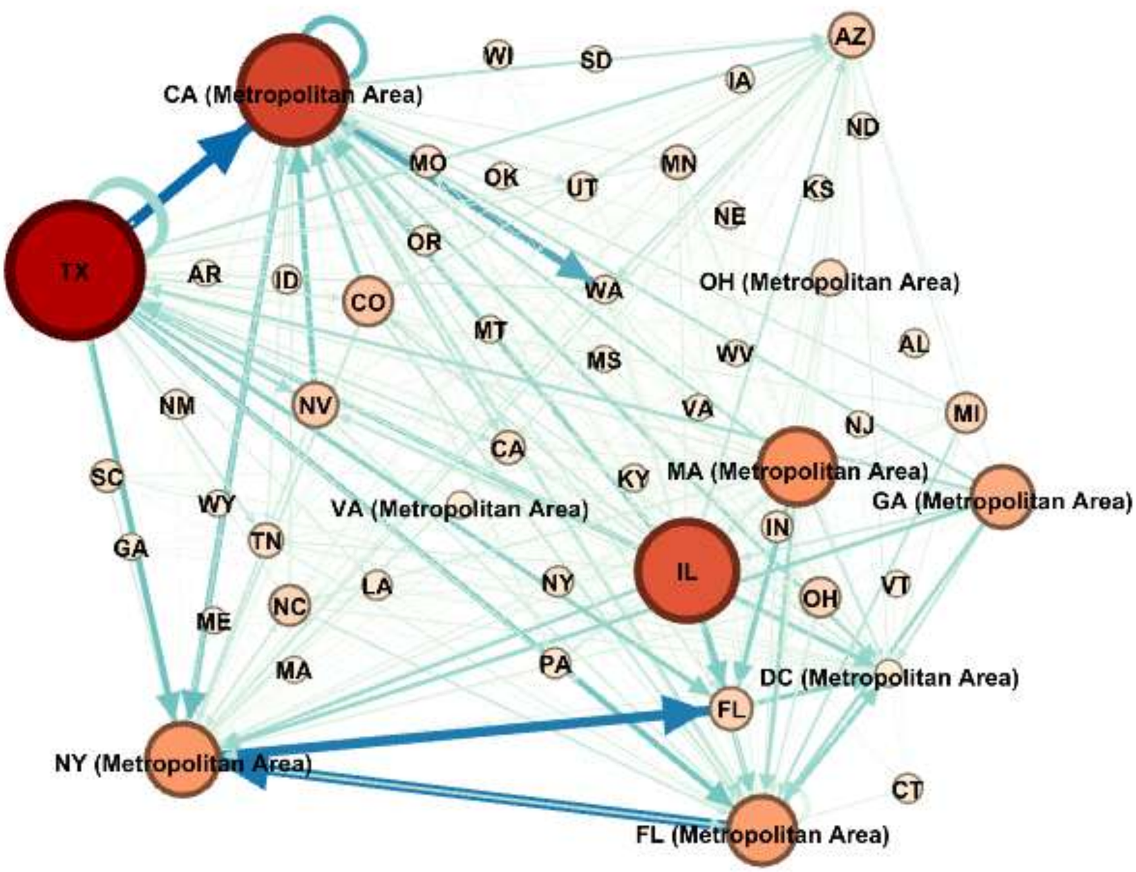
Id	Weighted In-Degree
CA (Metropolitan Area)	344435.0
NY (Metropolitan Area)	273921.0
FL (Metropolitan Area)	253575.0
DC (Metropolitan Area)	224458.0
FL	161407.0
AZ	137117.0
TX	115748.0
WA	82041.0
CA	59824.0
PA	46321.0



Note: Node's size and color Represent Weighted In–degree (passengers landing) or Weighted Out–degree (passengers departing)

2) passengers departing

Id	Weighted Out-Degree
TX	348198.0
CA (Metropolitan Area)	262969.0
IL	240787.0
MA (Metropolitan Area)	163966.0
NY (Metropolitan Area)	152645.0
FL (Metropolitan Area)	139066.0
GA (Metropolitan Area)	117943.0
CO	76119.0
NV	62393.0



Interpretation:

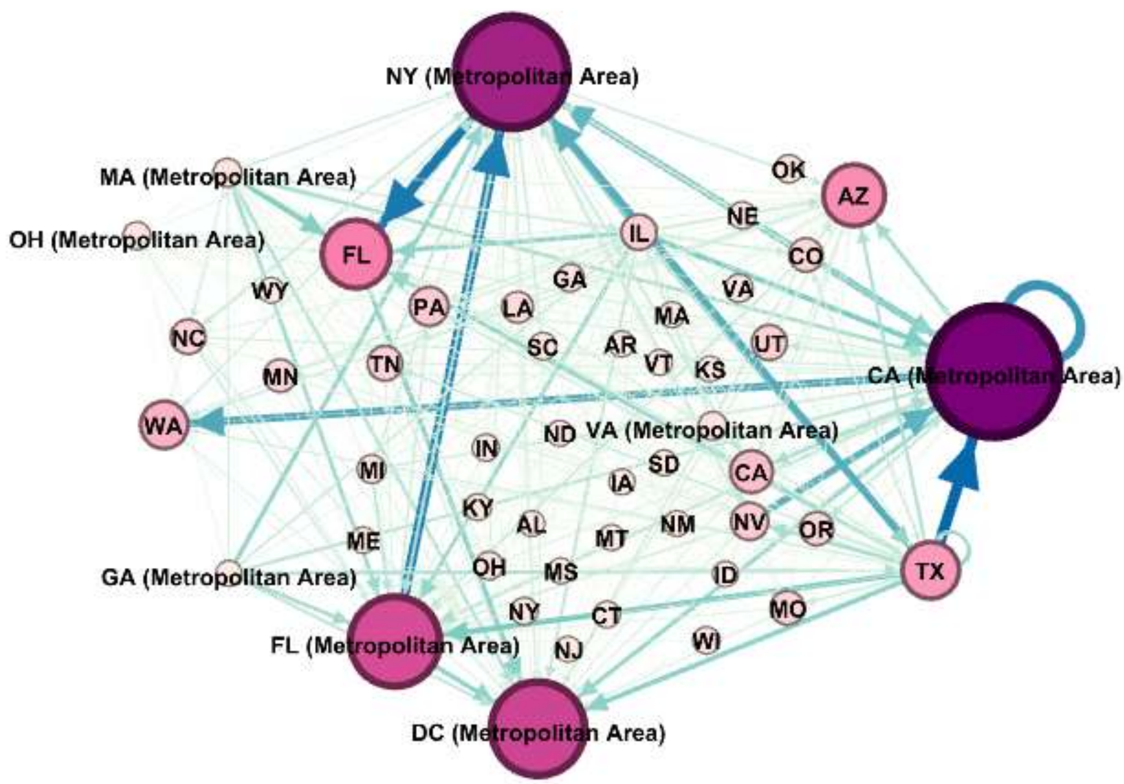
The 2021 network marks the beginning of the "re-consolidation" phase, where the traditional "Big Five" hubs began to reclaim their dominance. **California (CA)** and **New York (NY)** saw a resurgence in passenger volume, closing the gap with Texas and Florida. However, the recovery was uneven; the maps indicate that leisure-heavy destinations (like **Arizona** and **Nevada**) and the "Sun Belt" states continued to punch above their weight compared to 2019 levels. The network density increased, thickening the connections between these hubs as airlines restored frequencies.

Despite the recovery, the 2021 network retained a "leisure bias." The dominance of Florida remained disproportionately high compared to pre-pandemic years. The balance between In-Degree and Out-Degree stabilized, signaling that airlines had successfully reorganized their schedules to match the new, recovering demand patterns, but the overall volume had not yet returned to the saturation levels seen in 2019

Post−COVID−19 (Year 2022)

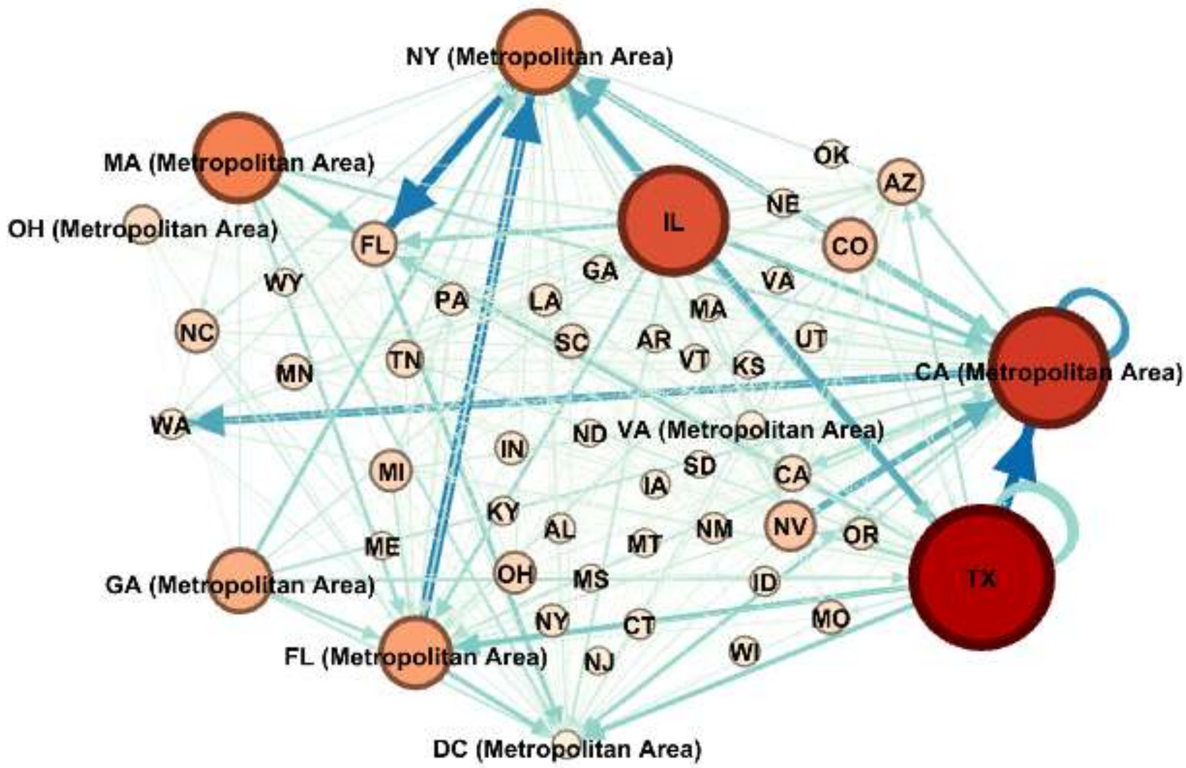
1) passengers landing

Id	Weighted In-Degree
CA (Metropolitan Area)	510047.0
NY (Metropolitan Area)	424591.0
DC (Metropolitan Area)	337078.0
FL (Metropolitan Area)	317682.0
FL	211569.0
AZ	178484.0
TX	155177.0
WA	109583.0
CA	83579.0



2) passengers departing

Id	Weighted Out-Degree
TX	467832.0
CA (Metropolitan Area)	371666.0
IL	330221.0
MA (Metropolitan Area)	253086.0
NY (Metropolitan Area)	230857.0
FL (Metropolitan Area)	183139.0
GA (Metropolitan Area)	152022.0
CO	108450.0
NV	94901.0



Interpretation:

By 2022, the network structure largely returned to its pre-pandemic topology, cementing the "Big Five" **(CA, TX, FL, NY, IL)** as the permanent anchors of the US aviation system. The visualization shows a robust, dense network where **New York** and **Illinois** have fully recovered their roles as critical connectors for the East and Midwest, balancing the persistent strength of Texas and Florida. The "passengers landing" and "departing" metrics are once again highly symmetrical, indicating a fully operational hub-and-spoke system running at high efficiency.

However, a key lasting change is visible: the network is now more consolidated around these super-nodes than ever before. Intermediate hubs play a supporting role, but the sheer volume of passengers flowing through the top 5 states suggests a "winner-take-all" dynamic. The resilience of the network now relies entirely on these massive hubs, as they serve as the dominant origin and destination for the vast majority of the country's air traffic.

Did it Change? Yes, but the change was in relative dominance rather than absolute replacement.

- The Shift: During the pandemic (2020), the hierarchy temporarily shuffled: New York's dominance plummeted, while Florida and Texas rose in importance due to resilient leisure travel.
- The Recovery: By 2022, the traditional hierarchy was largely restored, but with Florida retaining a permanently elevated status as a top-tier hub alongside California and Texas. The fundamental structure of the network proved resilient, returning to its core "Big Five" configuration after the systemic shock.

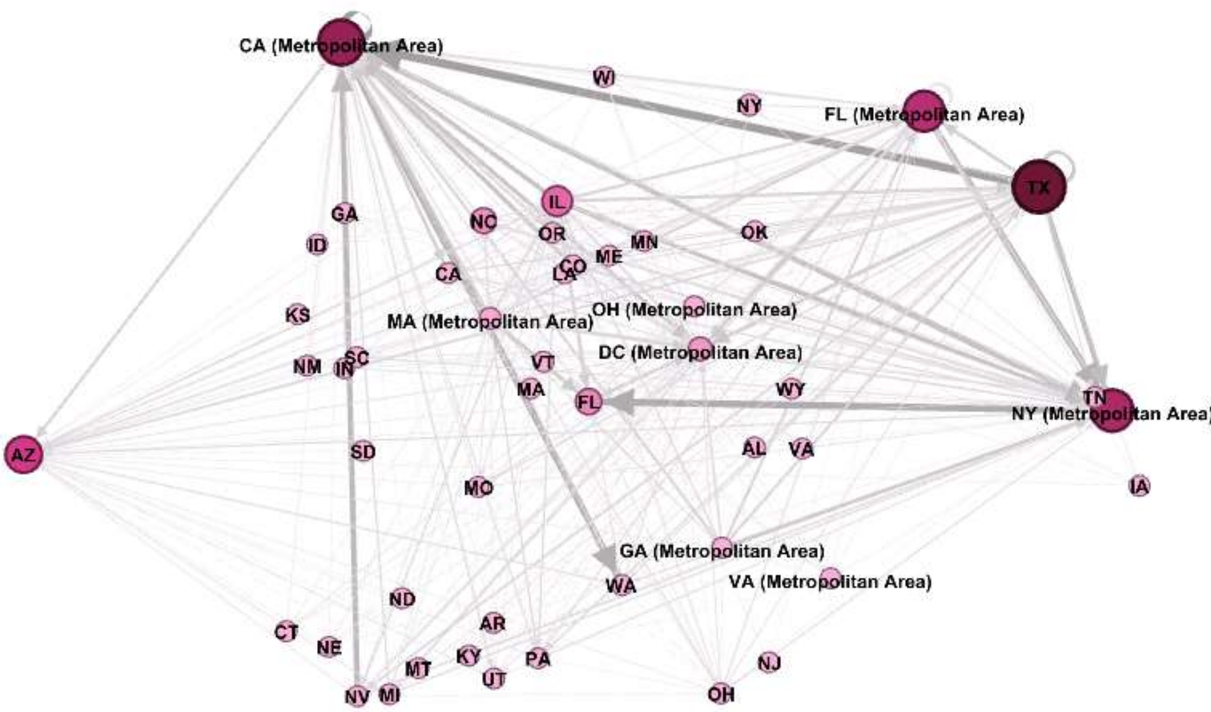
3) Which states are bottlenecks that most traffic must pass through?

To identify bottlenecks in the U.S. airline network, weighted betweenness centrality was computed for each state for the period 2019–2022. In this network, edge weights represent the number of passengers transported between states, allowing the analysis to account not only for network structure but also for the intensity of passenger flows.

Betweenness centrality measures the extent to which a state lies on the shortest passenger-weighted paths between other states and therefore identifies states through which a large share of passenger traffic must pass. Higher values indicate states that function as critical intermediaries in the movement of passengers across the network.

Pre-COVID-19 (Year 2019)

Id	Betweenness Centrality
TX	457.120563
CA (Metropolitan Area)	363.659668
NY (Metropolitan Area)	316.904113
FL (Metropolitan Area)	292.975144
AZ	235.060281
IL	141.211111
FL	82.371429
NC	73.52215
DC (Metropolitan Area)	46.791667
NY	15.168398
MA (Metropolitan Area)	12.532937
OH (Metropolitan Area)	3.804762
OH	2.595238
VA (Metropolitan Area)	2.4
NV	1.59127
MO	0.805556
CA	0.6
WA	0.6
CO	0.285714



Interpretation:

In 2019, the airline network exhibited a **highly centralized structure**. The network diameter was **5**, and the average path length was approximately **2.06**, indicating that most pairs of states were connected through only one or two intermediary states.

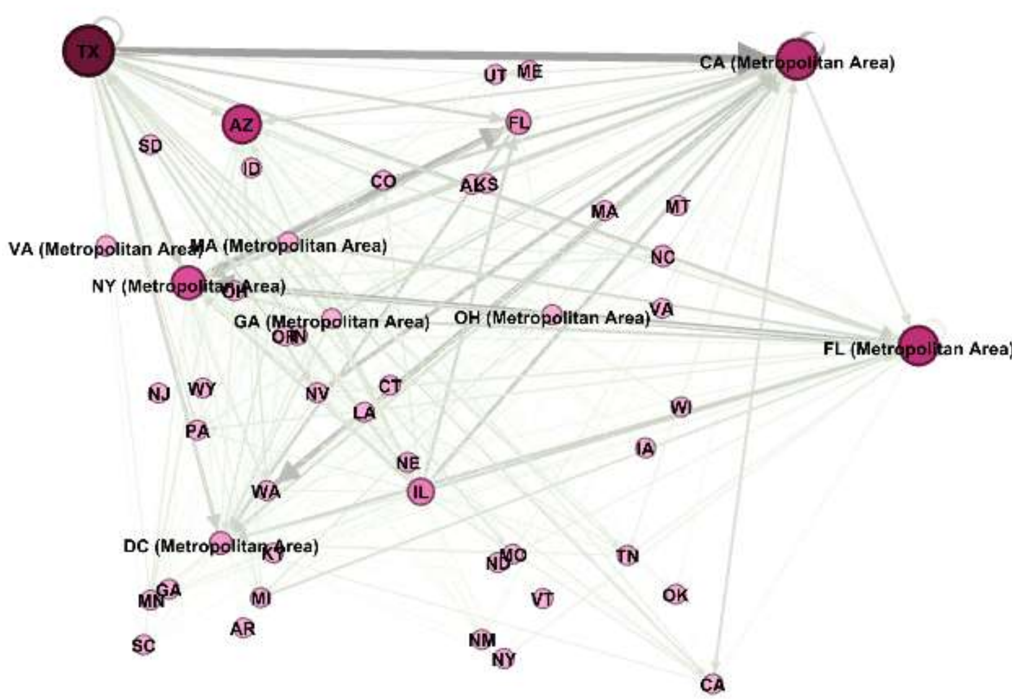
Texas emerged as the dominant bottleneck, showing the highest weighted betweenness centrality, which implies that a large proportion of passenger-weighted shortest paths passed through Texas.

Major metropolitan hubs—**California (Metropolitan Area)**, **New York (Metropolitan Area)**, and **Florida (Metropolitan Area)**—also played critical intermediary roles, facilitating the movement of large passenger volumes across regions.

These results indicate that prior to COVID-19, passenger traffic in the U.S. airline network was heavily concentrated through a small number of high-capacity hub states.

During COVID-19 (Year 2020)

Id	Betweenness Centrality
TX	518.630159
CA (Metropolitan Area)	355.039683
FL (Metropolitan Area)	333.278968
AZ	314.986111
NY (Metropolitan Area)	231.178968
IL	117.753571
FL	81.575
DC (Metropolitan Area)	50.091667
NC	27.768254
VA (Metropolitan Area)	8.777778
MA (Metropolitan Area)	3.854762
OH (Metropolitan Area)	3.688095
OH	2.751984
NY	2.394444



Interpretation:

In 2020, during the peak of COVID-19 disruptions, the network diameter decreased to $\mathbf{4}$, and the average path length slightly declined to $\mathbf{2.02}$, reflecting a contraction in network reach and route diversity.

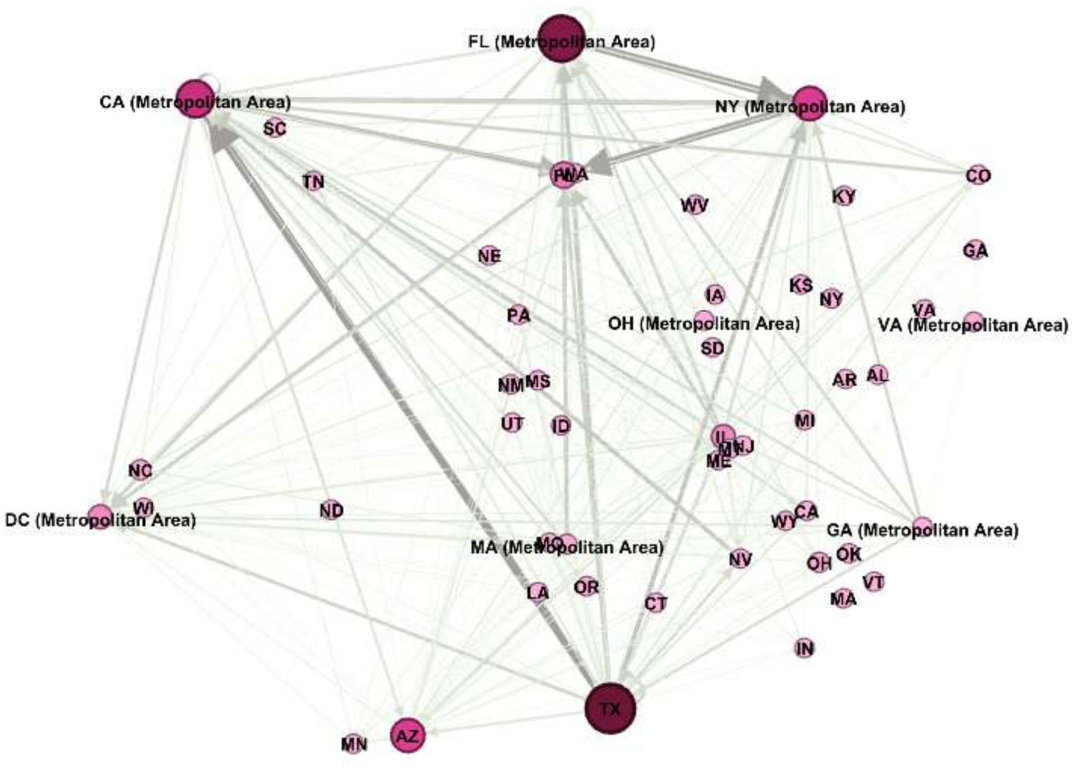
At the same time, weighted betweenness centrality became **more concentrated** among a few states. **Texas** further strengthened its position as the primary bottleneck for passenger flows, while **Florida (Metropolitan Area)** and **Arizona** gained relative importance.

In contrast, **New York (Metropolitan Area)** experienced a notable decline in betweenness, consistent with sharp reductions in passenger volumes. Many states displayed near-zero betweenness, indicating limited participation in connecting passenger flows.

Overall, the results suggest increased **network centralization and vulnerability**, with passenger traffic relying on fewer intermediary states during the pandemic.

Early Recovery (Year 2021)

Id	Betweenness Centrality
TX	577.840873
FL (Metropolitan Area)	518.762302
CA (Metropolitan Area)	349.325
NY (Metropolitan Area)	283.410317
AZ	277.994841
FL	133.620238
DC (Metropolitan Area)	91.95119
IL	89.577778
NC	19.990079
OH	15.315079
MA (Metropolitan Area)	11.29246
VA (Metropolitan Area)	3.761905
OH (Metropolitan Area)	3.526984
NY	3.241667
CA	2.027381
WA	1.017857
NV	0.917857
CO	0.767857
MT	0.533333



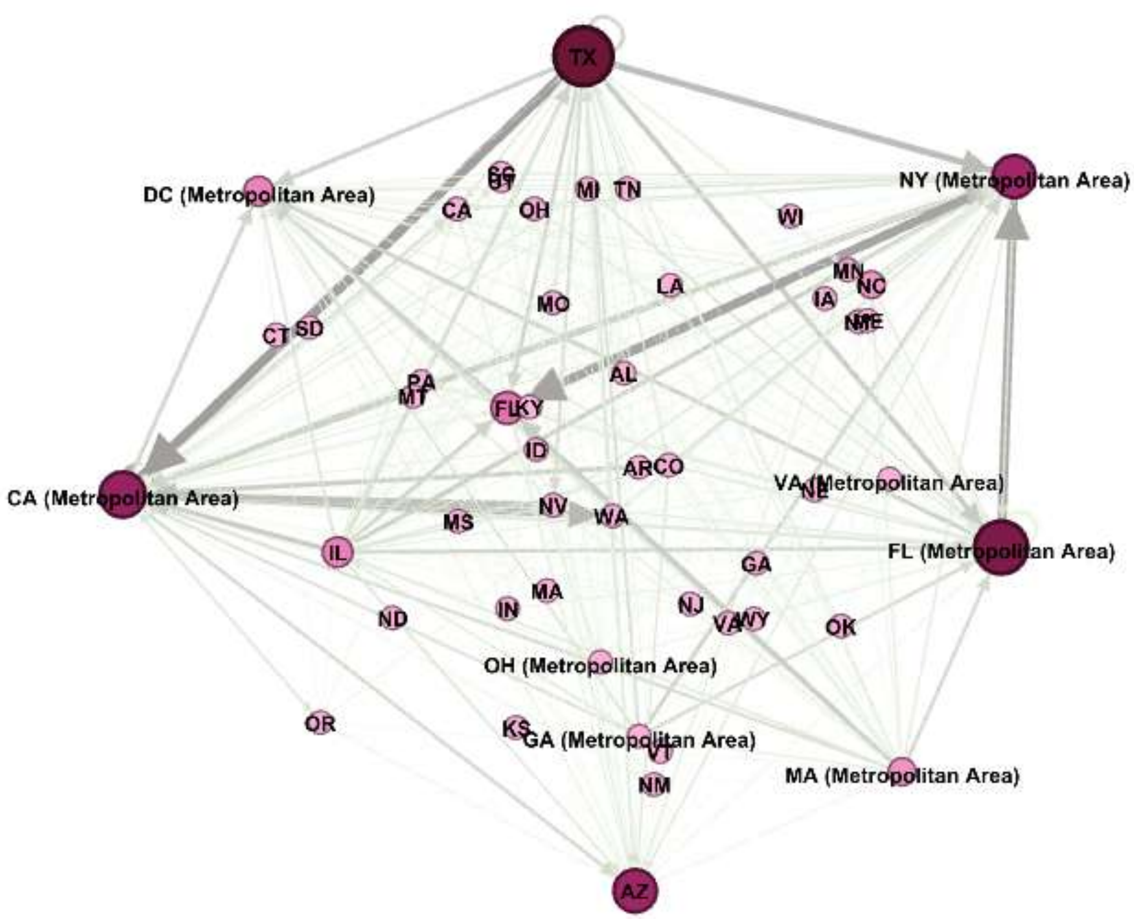
Interpretation:

By 2021, early signs of recovery were observed. The network diameter remained **4**, while the average path length increased slightly to **2.08**, indicating a gradual re-expansion of connectivity. **Texas** and **Florida (Metropolitan Area)** continued to dominate as passenger bottlenecks, mediating a substantial share of passenger-weighted shortest paths.

California (Metropolitan Area) and **New York (Metropolitan Area)** partially regained their intermediary roles as passenger demand recovered. The presence of moderate betweenness values in additional states suggests that passenger flows began to redistribute across a broader set of routes, although reliance on major hubs remained strong.

Post−COVID−19 (Year 2022)

Id	Betweenness Centrality
TX	505.980519
FL (Metropolitan Area)	426.159488
CA (Metropolitan Area)	319.076371
AZ	282.441053
NY (Metropolitan Area)	277.731421
FL	129.745635
IL	98.784524
DC (Metropolitan Area)	90.795635
MA (Metropolitan Area)	61.862302
NC	55.434885
NY	4.743038
OH (Metropolitan Area)	3.231349
SC	3.084019
NV	1.920635
CA	1.708333
WA	1.708333
VA (Metropolitan Area)	1.4



Interpretation:

In 2022, the airline network showed further stabilization. The network diameter increased again to 5, and the average path length rose slightly to 2.10, indicating improved connectivity and route availability compared to the pandemic period. Despite this recovery, bottlenecks persisted.

Texas remained the most important intermediary state, followed by Florida (Metropolitan Area), California (Metropolitan Area), Arizona, and New York (Metropolitan Area). While more states exhibited non-zero betweenness values, indicating a broader distribution of passenger flows, the overall structure did not fully return to its pre-COVID configuration.

Overall Interpretation:

Across 2019–2022, the U.S. airline network consistently exhibited a small-world structure, with short average path lengths and strong reliance on a limited number of bottleneck states. Texas emerged as the most persistent and dominant bottleneck throughout the period, lying on a disproportionately large number of passenger-weighted shortest paths.

The COVID-19 pandemic intensified network centralization, increasing dependence on a few key states to facilitate passenger movements. Although connectivity improved after the pandemic, the continued dominance of the same bottleneck states suggests that structural dependencies in the U.S. airline network remained largely unchanged.

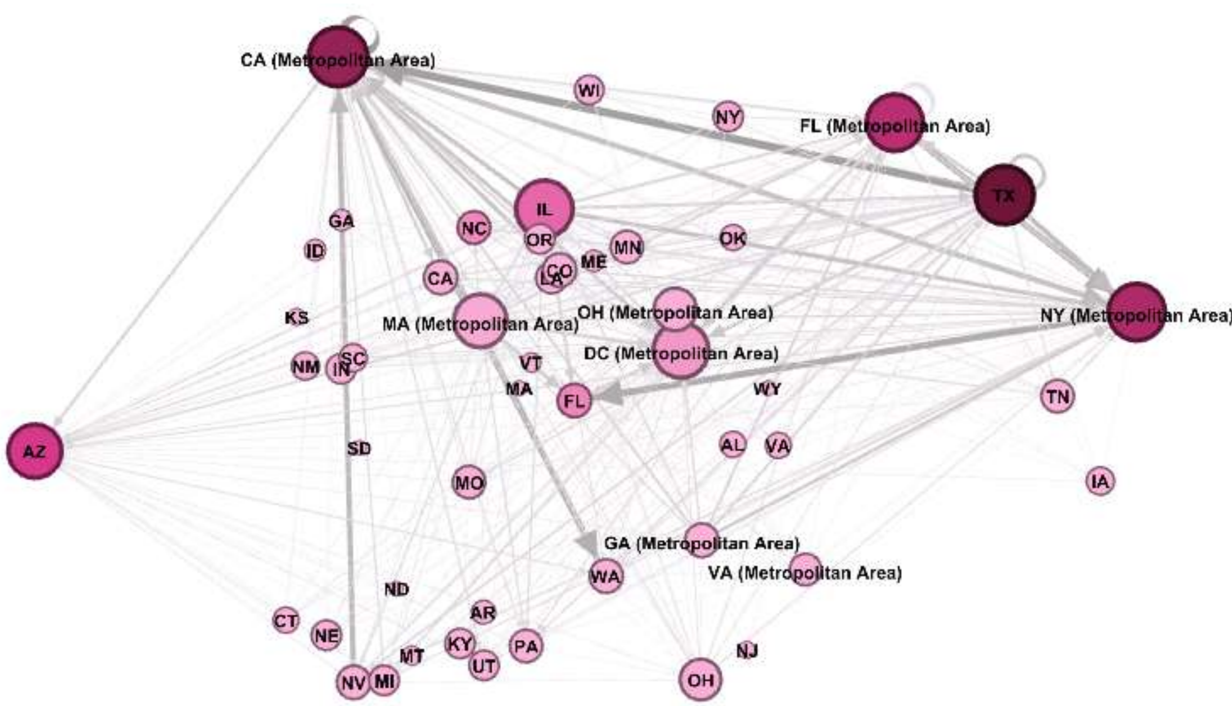
4)Which states serve as “air travel influencers” whose connectivity boosts nearby states?

To identify states that act as air travel influencers, **eigenvector centrality** was employed as it captures not only the extent of a state’s connectivity but also the importance of the states to which it is connected. This measure assigns higher values to states that are linked to other highly connected and influential states, thereby reflecting their ability to enhance the accessibility and network prominence of neighboring states.

Passenger volumes were incorporated as weights, ensuring that connections with higher passenger flows exert a greater influence on the centrality scores. By applying weighted eigenvector centrality for each year from 2019 to 2022, the analysis highlights state whose role in the airline system extends beyond direct traffic volumes to include their broader structural influence on the surrounding network.

Pre–COVID–19 (Year 2019)

Id	Eigenvector Centrality
CA (Metropolitan Area)	1.0
TX	0.987755
NY (Metropolitan Area)	0.955706
IL	0.954016
FL (Metropolitan Area)	0.950536
DC (Metropolitan Area)	0.924003
AZ	0.884426
MA (Metropolitan Area)	0.873959
OH (Metropolitan Area)	0.650078
OH	0.61256
CA	0.454274
WA	0.454274
FL	0.454274
CO	0.454274
NV	0.454274
GA (Metropolitan Area)	0.454274
NC	0.433705
MO	0.433705
PA	0.433705
MN	0.433705



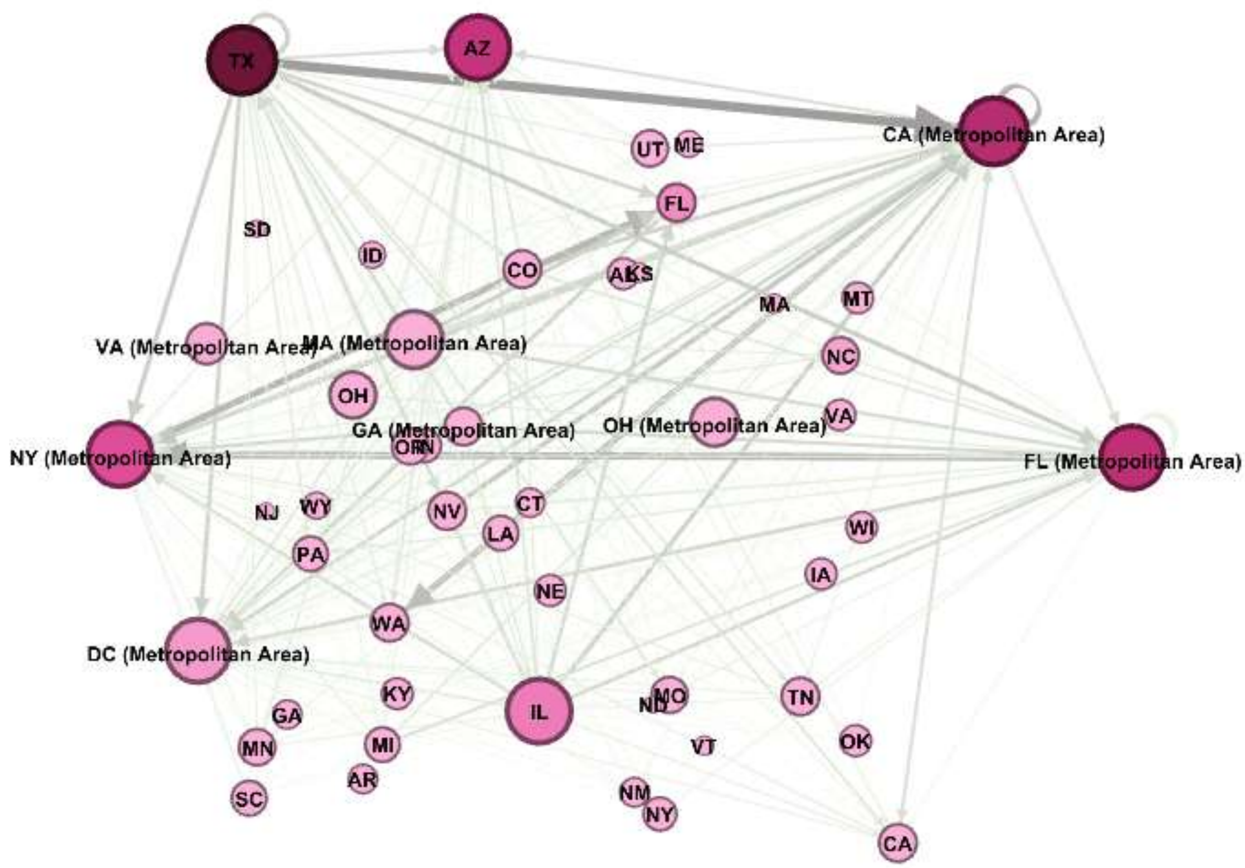
Interpretation:

In 2019, the airline network displayed a **strongly interconnected core** dominated by major metropolitan hubs. **California (Metropolitan Area)** recorded the highest eigenvector centrality, closely followed by **Texas**, **New York (Metropolitan Area)**, **Illinois**, and **Florida (Metropolitan Area)**.

High eigenvector centrality values for these states indicate that they were not only highly connected but also connected to other highly connected states through high-volume passenger routes. As a result, increases in connectivity in these hubs likely generated substantial spillover benefits for nearby and connected states, reinforcing their role as influential centers in the national airline network.

During COVID−19 (Year 2020)

Id	Eigenvector Centrality
TX	1.0
CA (Metropolitan Area)	0.993763
IL	0.940094
NY (Metropolitan Area)	0.939982
AZ	0.93021
DC (Metropolitan Area)	0.926234
FL (Metropolitan Area)	0.925761
MA (Metropolitan Area)	0.811959
OH (Metropolitan Area)	0.639222
OH	0.601048
VA (Metropolitan Area)	0.496776
CA	0.439477
WA	0.439477
NV	0.439477
TN	0.439477



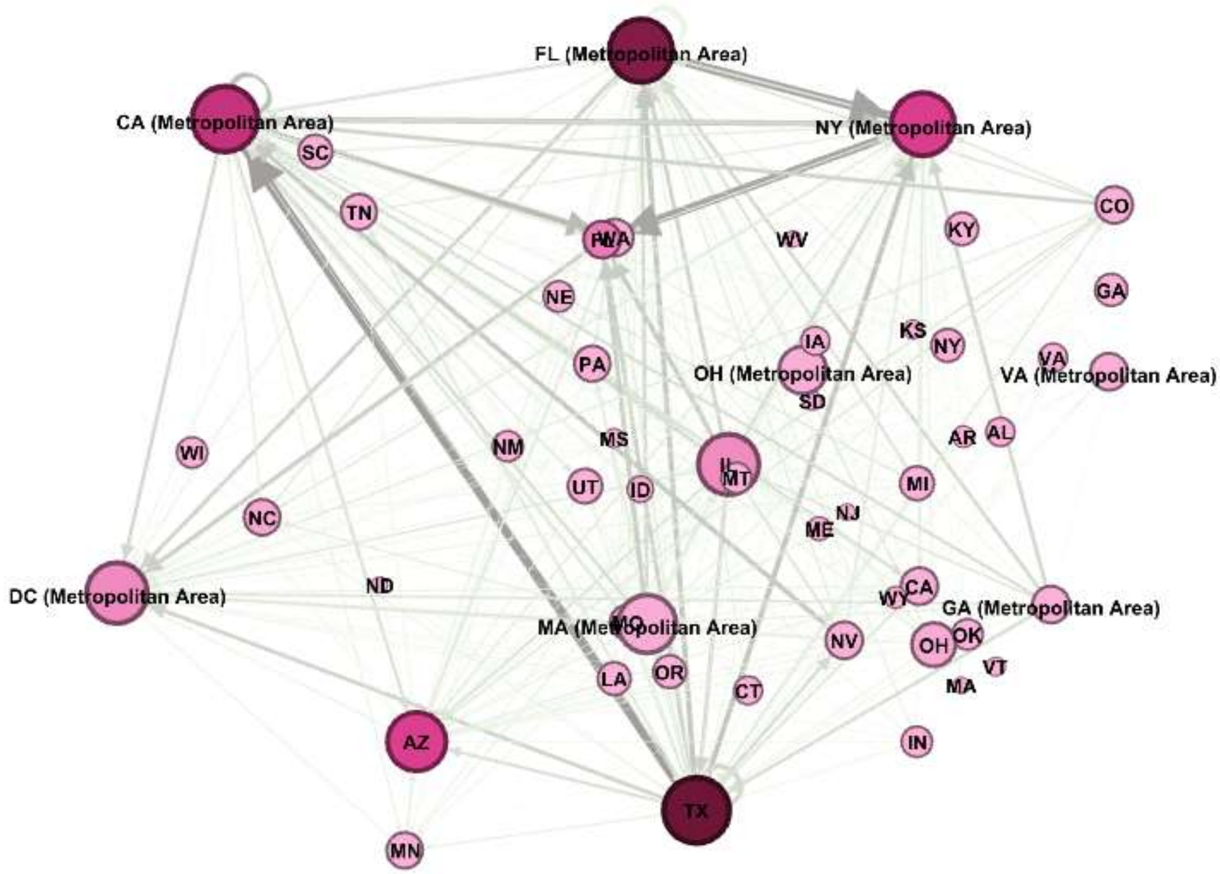
Interpretation:

In 2020, the pandemic significantly disrupted passenger volumes and network structure. Despite these shocks, Texas emerged as the most influential state, achieving the highest eigenvector centrality. California (Metropolitan Area), Illinois, and New York (Metropolitan Area) remained highly ranked, although with slightly reduced relative influence.

The persistence of these states at the top of the ranking suggests that even under severe demand contraction, a small group of core hubs continued to support connectivity for surrounding states. Meanwhile, many peripheral states experienced low eigenvector centrality, indicating diminished indirect benefits from the network during the pandemic.

Early Recovery (Year 2021)

Id	Eigenvector Centrality
TX	1.0
CA (Metropolitan Area)	0.999907
FL (Metropolitan Area)	0.957162
NY (Metropolitan Area)	0.951579
DC (Metropolitan Area)	0.90807
IL	0.900563
AZ	0.865201
MA (Metropolitan Area)	0.854743
OH (Metropolitan Area)	0.649324
OH	0.573289
VA (Metropolitan Area)	0.445417
NV	0.444365
FL	0.444365
CO	0.444365
GA (Metropolitan Area)	0.444365



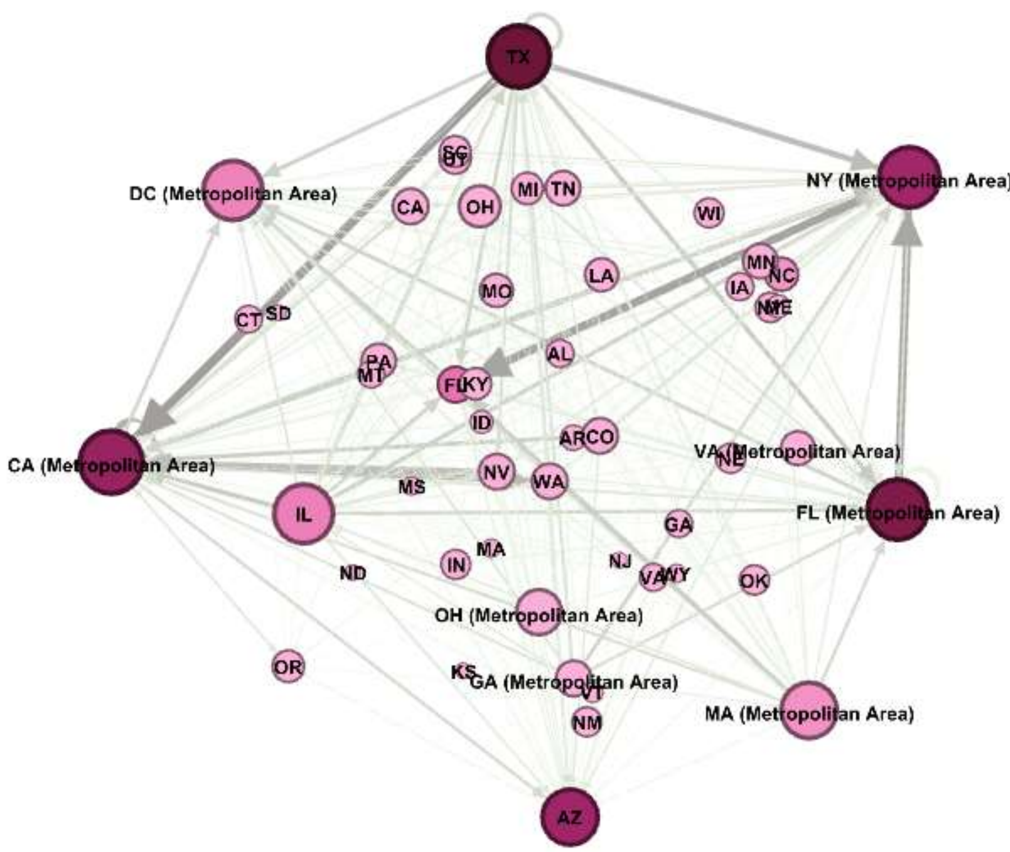
Interpretation:

By 2021, as passenger demand began to recover, eigenvector centrality values reflected a partial rebalancing of influence within the network. **Texas** and **California (Metropolitan Area)** remained dominant influencers, while **Florida (Metropolitan Area)** and **New York (Metropolitan Area)** regained prominence, consistent with the rebound in domestic leisure and business travel.

The increasing influence of these hubs suggests that improvements in their connectivity played a critical role in restoring accessibility and passenger flows for neighboring and connected states during the recovery phase.

Post−COVID−19 (Year 2022)

Id	Eigenvector Centrality
CA (Metropolitan Area)	1.0
TX	0.98627
NY (Metropolitan Area)	0.951111
FL (Metropolitan Area)	0.947174
DC (Metropolitan Area)	0.926795
IL	0.919879
MA (Metropolitan Area)	0.856786
AZ	0.851996
OH (Metropolitan Area)	0.640734
OH	0.563826
NV	0.447788
CA	0.447788
WA	0.447788
FL	0.447788
CO	0.447788
GA (Metropolitan Area)	0.447788
PA	0.427386
MN	0.427386
TN	0.427386



Interpretation:

In 2022, the airline network exhibited further stabilization. **California** returned to the top position in eigenvector centrality, followed closely by **Texas**, **New York (Metropolitan Area)**, and **Florida (Metropolitan Area)**.

These results indicate a re–established influential core similar to the pre–pandemic structure. States connected to these major hubs benefited most from renewed passenger volumes and network density, highlighting the continued importance of large metropolitan areas as connectivity multipliers in the post–COVID airline system.

Overall Interpretation:

Across **2019–2022**, eigenvector centrality reveals a **persistent hierarchy of influential states** within the U.S. airline network. **California (Metropolitan Area)**, **Texas**, **New York (Metropolitan Area)**, and **Florida (Metropolitan Area)** consistently functioned as air travel influencers whose connectivity amplified the importance and accessibility of surrounding states. Although COVID–19 temporarily altered passenger volumes and shifted relative influence among hubs, the overall structure of influence remained remarkably stable.

This suggests that the pandemic disrupted traffic levels more than it transformed the underlying connectivity dynamics of the U.S. airline network, states suggests that structural dependencies in the U.S. airline network remained largely unchanged.

5)Which states are most influential in spreading disruptions or delays?

(If a delay starts in one state, which state is most likely to spread the disruption widely?)

To address this research question, we need to study the closeness of each state:

2019		2020	
State	Closeness	State	Closeness
IL	0.909091	TX	0.933333
TX	0.851064	IL	0.875
MA (Metropolitan Area)	0.833333	MA (Metropolitan Area)	0.792453
CA (Metropolitan Area)	0.714286	CA (Metropolitan Area)	0.736842
FL (Metropolitan Area)	0.666667	FL (Metropolitan Area)	0.677419
NY (Metropolitan Area)	0.666667	NY (Metropolitan Area)	0.666667
OH (Metropolitan Area)	0.634921	AZ	0.636364
OH	0.615385	OH (Metropolitan Area)	0.636364
AZ	0.588235	OH	0.617647
GA (Metropolitan Area)	0.577465	GA (Metropolitan Area)	0.573333

2021		2022	
State	Closeness	State	Closeness
TX	0.895833	TX	0.875
IL	0.796296	IL	0.807692
MA (Metropolitan Area)	0.796296	MA (Metropolitan Area)	0.807692
CA (Metropolitan Area)	0.728814	FL (Metropolitan Area)	0.7
FL (Metropolitan Area)	0.704918	CA (Metropolitan Area)	0.688525
NY (Metropolitan Area)	0.671875	NY (Metropolitan Area)	0.65625
OH (Metropolitan Area)	0.632353	OH (Metropolitan Area)	0.617647
AZ	0.623188	AZ	0.591549
OH	0.597222	OH	0.591549
GA (Metropolitan Area)	0.571429	GA (Metropolitan Area)	0.573333

Interpretation:

Based on **closeness centrality**, **Texas, Illinois, Massachusetts (Metropolitan Area), California (Metropolitan Area), and Florida (Metropolitan Area)** consistently emerge as the most influential states in the US airline network in terms of the potential spread of disruptions or delays.

As shown in the table, these states record the **highest closeness values across the four-year period**, with values generally ranging between **0.70 and 0.93**. For example, **Texas** attains the highest closeness in **2020 (0.933)** and remains highly central in **2021 (0.896)** and **2022 (0.875)**, while **Illinois** records similarly elevated values, including **0.909 in 2019** and **0.875 in 2020**. **Massachusetts (Metropolitan Area)**, **California (Metropolitan Area)**, and **Florida (Metropolitan Area)** consistently follow with closeness values between approximately **0.67 and 0.81**, confirming their structural proximity to the rest of the network.

A clear temporal pattern is also evident. In **2020**, closeness centrality values increase for the most central states—such as Texas and Illinois—reflecting the **contraction of the airline network during the COVID-19 pandemic.** With fewer active routes, the average shortest path lengths decreased, resulting in higher closeness scores despite the overall reduction in passenger traffic.

In **2021 and 2022**, closeness values gradually stabilize and move closer to pre-pandemic levels as routes were restored and connectivity improved. For instance, Texas' closeness declines from **0.933 in 2020** to **0.896 in 2021** and **0.875 in 2022**, while Illinois follows a similar adjustment pattern. Importantly, the **ranking of the most central states remains largely unchanged**, indicating that their structural importance within the network persisted throughout the disruption and recovery phases.

Overall interpretation:

The table and the network visualization jointly confirm that a small group of structurally central states consistently plays a dominant role in shaping how disruptions or delays propagate across the US airline network over time.

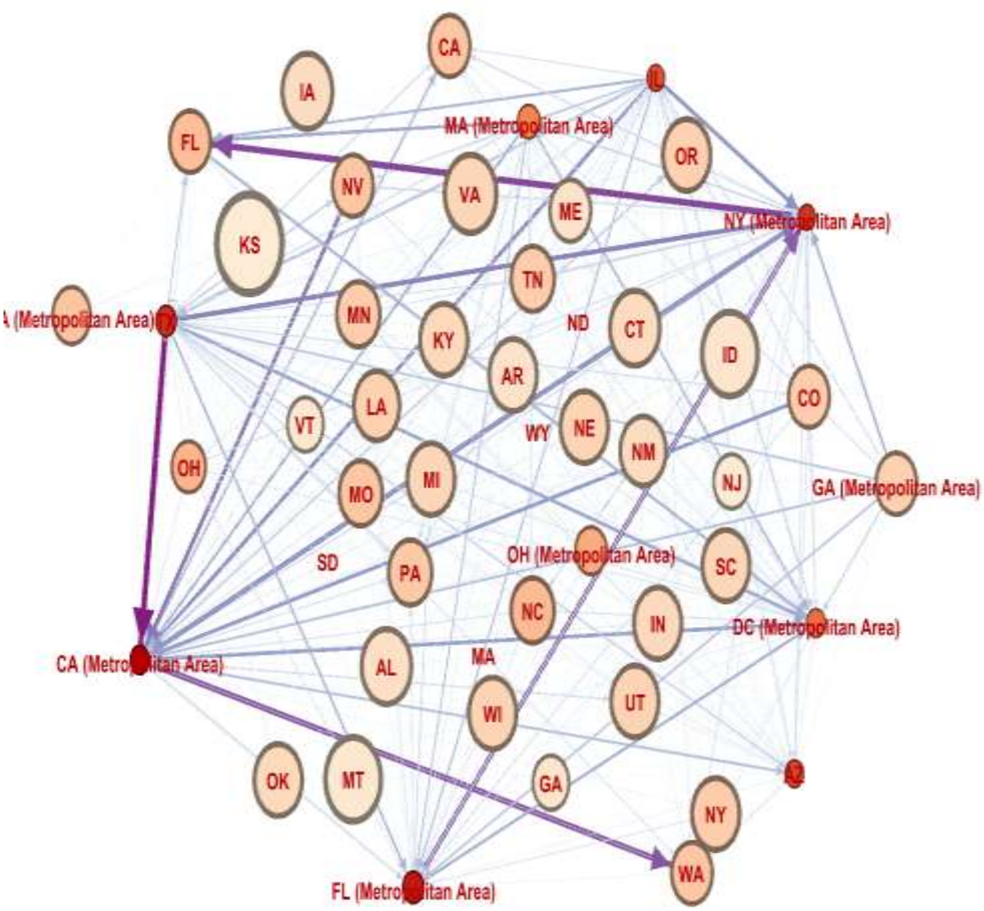
6) Which states maximize short-range regional connectivity?

Short-range regional connectivity reflects how well a state connects its immediate neighbors into tightly-knit clusters. States that act as strong regional connectors form dense local triangles, creating cohesive local airline networks. These states play a crucial role in maintaining network stability and regional accessibility, especially during periods of disruption such as the COVID-19 pandemic.

In this analysis, the clustering coefficient is used to identify states that maximize short-range regional connectivity. By examining clustering patterns from 2019 to 2022, the analysis highlights how local connectivity structures evolved before, during, and after the pandemic, and whether regional airline networks remained resilient despite major shocks to national and international travel.

Pre–COVID–19 (Year 2019)

Id	clustering
KS	1
ID	0.833333
MT	0.833333
VA	0.766667
IA	0.738095
AL	0.733333
CT	0.733333
AR	0.7
OK	0.7



Interpretation:

The 2019 clustering results reveal a clear distinction between states that maximize short-range regional connectivity and those that primarily function as national or metropolitan hubs.

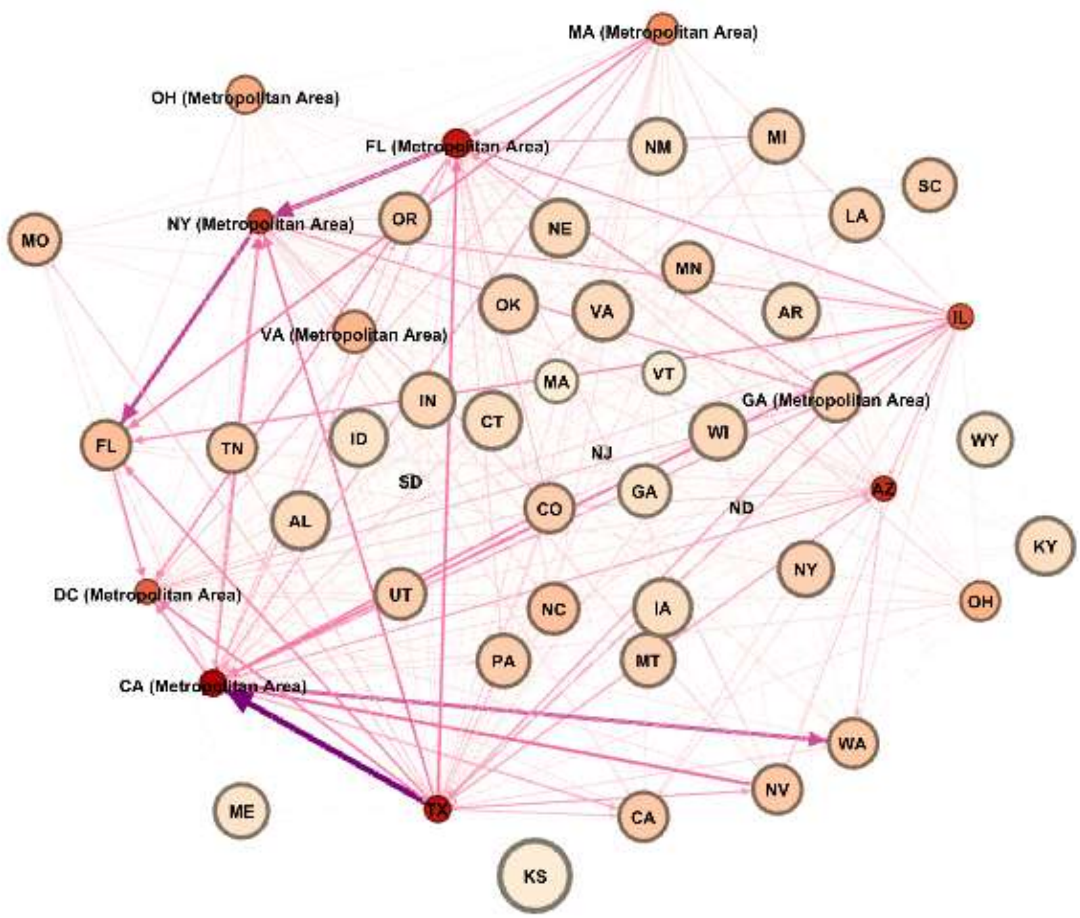
States with high clustering coefficients—such as **Kansas, Idaho, Montana, Virginia, Iowa, Alabama, and Connecticut**—exhibit tightly knit local airline networks, where neighboring states are strongly interconnected. These dense local structures indicate that air travel within these regions is supported by cohesive short-distance routes rather than reliance on long-haul connections. Such states effectively act as regional connectors, reinforcing accessibility and stability within their immediate geographic areas.

In contrast, major metropolitan hubs—including **Texas and large metropolitan areas in California, Florida, and New York**—exhibit low clustering, reflecting their role in facilitating long-distance and national travel rather than strengthening local connectivity.

Overall, the results demonstrate that regional airline stability in 2019 relied on cohesive local networks rather than dominant national hubs, providing a strong foundation for resilience in later years.

During COVID-19 (Year 2020)

Id	clustering
KS	1
VA	0.785714
CT	0.766667
AL	0.761905
OK	0.761905
KY	0.761905
NE	0.761905
IA	0.761905
ID	0.75
WI	0.738095
AR	0.733333
NM	0.733333
IN	0.714286
NY	0.714286
ME	0.7
WY	0.7



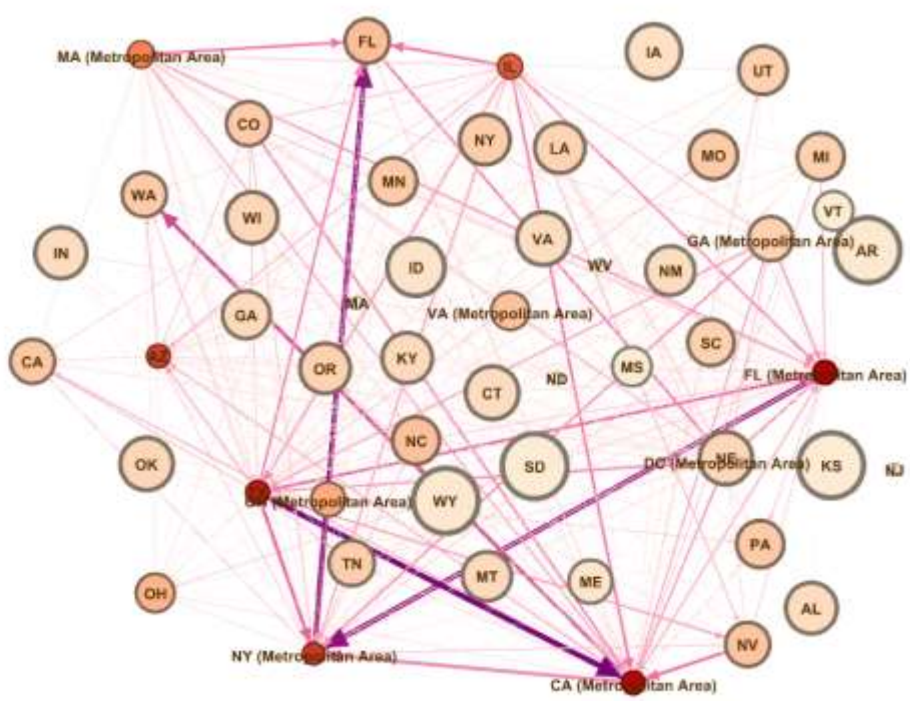
Interpretation:

In 2020, the U.S. airline network became more regionally focused, with many states showing high clustering coefficients, indicating strong local connectivity. Kansas (KS), Virginia (VA), Connecticut (CT), Alabama (AL), Oklahoma (OK), Kentucky (KY), Nebraska (NE), Iowa (IA), Idaho (ID), and Wisconsin (WI) emerged as key regional stabilizers, maintaining dense short-range connections even as national and long-distance travel declined. This shift highlights how regional networks played a crucial role in preserving mobility during pandemic-related disruptions.

On the other hand, large metropolitan areas and traditionally dominant hubs show much lower clustering. This indicates that hub-and-spoke structures weakened in 2020, as many long-haul and connecting flights were canceled. The network therefore shifted away from centralized connectivity toward localized, resilient regional structures.

Early Recovery (Year 2021)

Id	clustering
AR	1
KS	1
SD	1
WY	1
ID	0.85
IA	0.8
VA	0.766667
CT	0.766667
OK	0.761905
NE	0.761905
WI	0.738095
IN	0.738095
AL	0.733333
OR	0.714286
KY	0.714286
LA	0.714286
NY	0.714286



Interpretation:

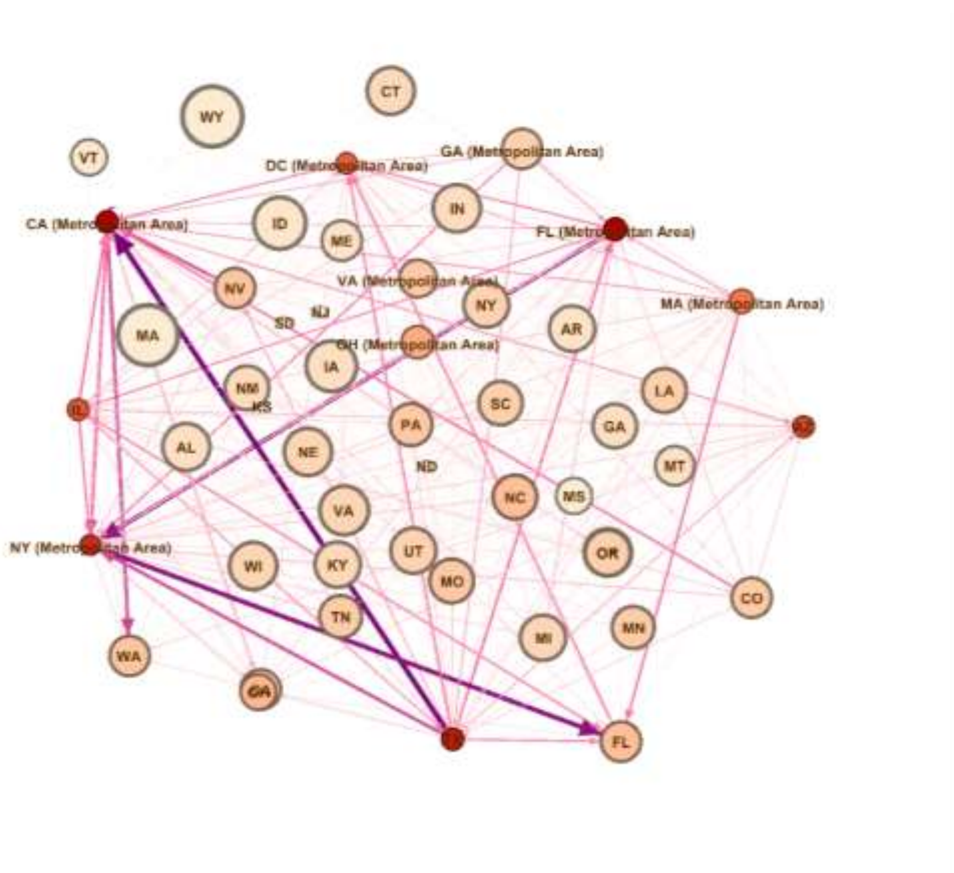
In 2021, the U.S. airline network maintained strong short-range regional connectivity, similar to previous years. States with the highest clustering coefficients—including Arkansas (AR), Kansas (KS), South Dakota (SD), Wyoming (WY), Idaho (ID), Iowa (IA), Virginia (VA), Connecticut (CT), Oklahoma (OK), Nebraska (NE), Wisconsin (WI), Kentucky (KY), Oregon (OR), New York (NY), Indiana (IN), New Mexico (NM), Utah (UT), and Michigan (MI)—acted as regional hubs, forming dense local travel triangles and sustaining mobility within nearby regions.

These states ensured that even when national hubs were affected by fluctuations in travel demand, local connectivity remained resilient. States like Massachusetts (MA), North Dakota (ND), New Jersey (NJ), and West Virginia (WV) showed very low or zero clustering, reflecting minimal regional connections and a limited role in local networks.

Overall, the 2021 results show a continuation of the regionalized pattern observed during the COVID-19 pandemic, emphasizing the importance of states with dense local networks in preserving short-range travel stability.

Post−COVID−19 (Year 2022)

Id	clustering
MA	1
WY	1
ID	0.833333
IA	0.8
VA	0.766667
OK	0.738095
NE	0.738095
WI	0.738095
IN	0.738095
AL	0.733333
CT	0.733333
AR	0.7



Interpretation:

In 2022, the U.S. airline network maintained strong regional connectivity, with states like Massachusetts (MA), Wyoming (WY), Idaho (ID), and Iowa (IA) showing the highest clustering, reflecting dense local connections. These states acted as regional hubs, sustaining short-range mobility despite ongoing recovery from COVID-19 disruptions.

States like Ohio (OH), Mississippi (MS), Vermont (VT), and Virginia (Metropolitan Area) showed moderate clustering, indicating partial regional connectivity, while Kansas (KS), South Dakota (SD), North Dakota (ND), and New Jersey (NJ) had zero clustering, suggesting minimal local interconnections.

Overall Interpretation:

The analysis of clustering coefficients from 2019 to 2022 shows that short-range regional connectivity plays a fundamental role in the resilience of the U.S. airline network. Before COVID-19, states with high clustering coefficients (such as Kansas, Idaho, Montana, and Virginia) already supported tightly knit local airline networks, indicating that regional stability relied more on cohesive short-distance connections than on dominant national hubs. Major metropolitan states, by contrast, primarily facilitated long-haul travel and exhibited lower local clustering.

During the COVID-19 pandemic in 2020, this pattern became even more pronounced. As long-distance and hub-based travel sharply declined, the airline network shifted toward regionalization. States with dense local connections emerged as critical stabilizers, maintaining mobility and accessibility despite severe disruptions. This highlights how regional airline structures provided a buffer against systemic shocks when centralized hub-and-spoke systems weakened.

The early recovery period in 2021 and the post-COVID phase in 2022 demonstrate that this regionalized structure largely persisted. Many states continued to exhibit high clustering, reinforcing the importance of localized networks in sustaining short-range travel even as national demand gradually returned. Although some states showed declining or zero clustering in 2022, the overall pattern confirms that regional connectors remain essential for network robustness.

Overall, the findings suggest that a resilient airline network depends not only on major national hubs but also (crucially) on states that maximize short-range regional connectivity. Dense local airline networks enhance stability, ensure continued accessibility during crises, and play a lasting role in the structure and recovery of the U.S. airline system.

7) Do major hubs in the U.S. airline network tend to connect mostly with other hubs or with smaller states?

Degree Assortativity:

Degree assortativity measures whether states with similar connectivity tend to connect to each other. Its value ranges from −1 to +1:

- Positive values: Highly connected states (major hubs) tend to link with other hubs.
- Negative values: Hubs mainly connect to smaller, less-connected states (typical hub-and-spoke).
- Near zero: No clear pattern in connections.

In the U.S. airline network, degree assortativity reveals the overall pattern of connections between states, rather than focusing on individual states. It helps identify whether the network is primarily centered around major hubs or if hubs tend to connect with each other, forming a core network.

- **Degree Assortativity of the network 2019: −0.630244155023758**
- **Degree Assortativity of the network 2020: −0.6379193738063432**
- **Degree Assortativity of the network 2021: −0.6340815547206409**
- **Degree Assortativity of the network 2022: −0.6321444658572051**

Interpretation:

From 2019 to 2022, the U.S. domestic airline network consistently exhibits a disassortative structure with degree assortativity values around -0.63 to -0.64.

This indicates a persistent hub-and-spoke pattern, where highly connected states (major hubs such as DC, NJ, MA, and CA) primarily connect with smaller or less-connected states.

The network shows stability in its global connectivity patterns, with major hubs playing a central role in maintaining the system, while smaller states remain dependent on these hubs. This complements the PFDI and Diversity results, highlighting both node-level dependency and network-level structure, and provides a comprehensive picture of U.S. airline network resilience and connectivity.

8) Passenger Flow Dependency and Diversity

Passenger Flow Dependency Index (PFDI):

- Measures how dependent a state is on a limited number of outgoing air travel connections.
- Higher values indicate concentration on few routes and higher vulnerability.

Passenger Flow Diversity Index (PFDI-D):

- Measures how diversified a state's outgoing connections are.
- Higher values indicate well-distributed flows and higher resilience.

Calculation:

For each state i, if w_{ij} is the number of passengers from state i to state j, the indices are calculated as:

$$\mathrm{PFDI}_i = \sum_j \left(\frac{w_{ij}}{\sum_k w_{ik}} \right)^2$$

PFDI−Di=1−PFDIi

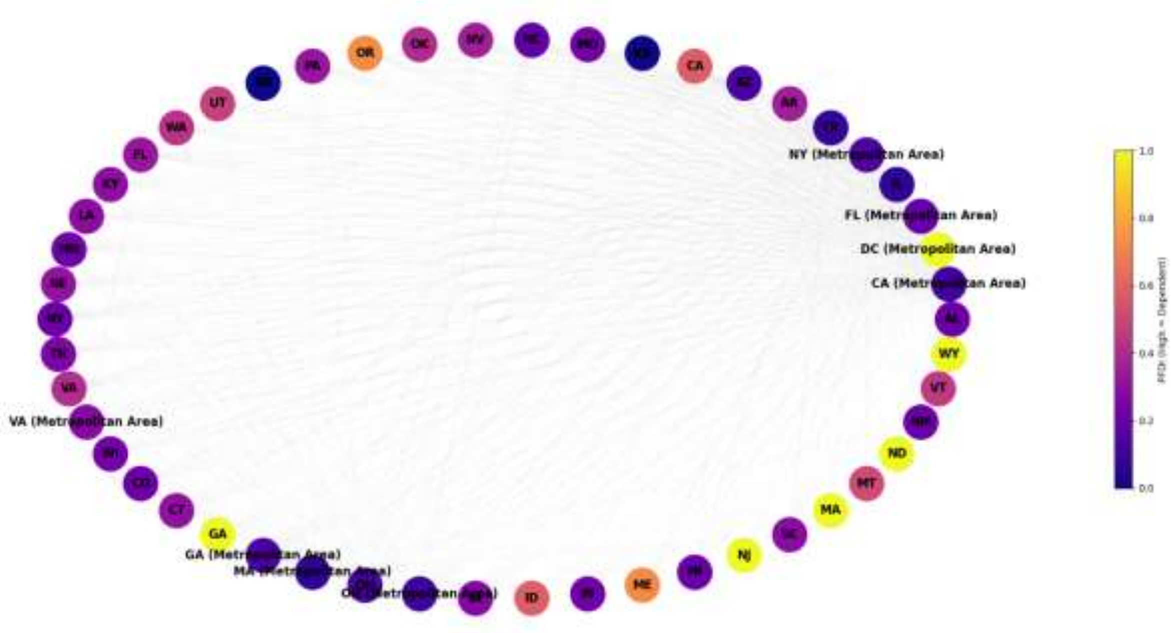
Where:

- w_{ij}= number of passengers traveling from state i to state j.
- $\sum_k w_{ik}$= total outgoing passengers from state i.

Results Interpretation:

Pre−COVID−19 (Year 2019)

	State	PFDI	Diversity
2	DC (Metropolitan Area)	1.000000	0.000000
32	GA	1.000000	0.000000
46	ND	1.000000	0.000000
49	WY	1.000000	0.000000
44	MA	1.000000	0.000000
42	NJ	1.000000	0.000000
15	OR	0.727946	0.272054
40	ME	0.721357	0.278643
38	ID	0.582961	0.417039
9	CA	0.580115	0.419885



States with PFDI = 1.0

- DC, GA, NJ, MA, ND, WY
- These states show **complete dependence on a single outgoing route**.
- Diversity = 0 confirms **no alternative paths**, making these states **highly fragile hubs** in the network.

States with moderate PFDI (0.58–0.73)

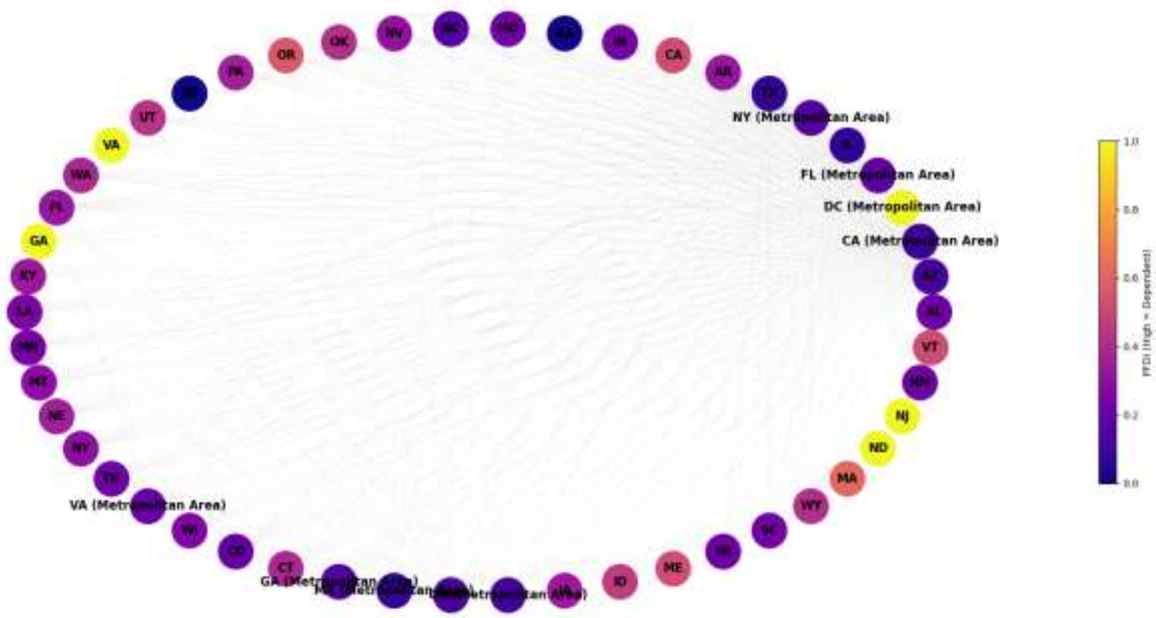
- OR (0.728), ME (0.721), ID (0.583), CA (0.58)
- These states have **some level of diversification**, with passenger flows distributed across a few routes.
- Diversity values between 0.27–0.42 indicate **moderate resilience**, able to withstand some disruptions.

Interpretation:

In 2019, the PFDI results reveal a clear contrast in the U.S. airline network: several small or metropolitan states are entirely dependent on a single route, while major hubs like California and Idaho show moderate diversification, indicating early structural resilience in parts of the network.

During COVID-19 (Year 2020)

	State	PFDI	Diversity
3	DC (Metropolitan Area)	1.000000	0.000000
20	VA	1.000000	0.000000
46	ND	1.000000	0.000000
47	NJ	1.000000	0.000000
23	GA	1.000000	0.000000
45	MA	0.616952	0.383048
16	OR	0.561019	0.438981
41	ME	0.520081	0.479919
9	CA	0.517643	0.482357
49	VT	0.516491	0.483509



States with PFDI = 1.0

- DC, VA, ND, NJ, GA
- These states have **all outgoing passengers concentrated on a single route**, meaning **complete dependency**.
- Diversity = 0 confirms **no alternative connections**, making these states **extremely fragile** in the network.

States with moderate PFDI (0.5–0.62)

- MA (0.616), OR (0.561), ME (0.520), CA (0.518), VT (0.516)

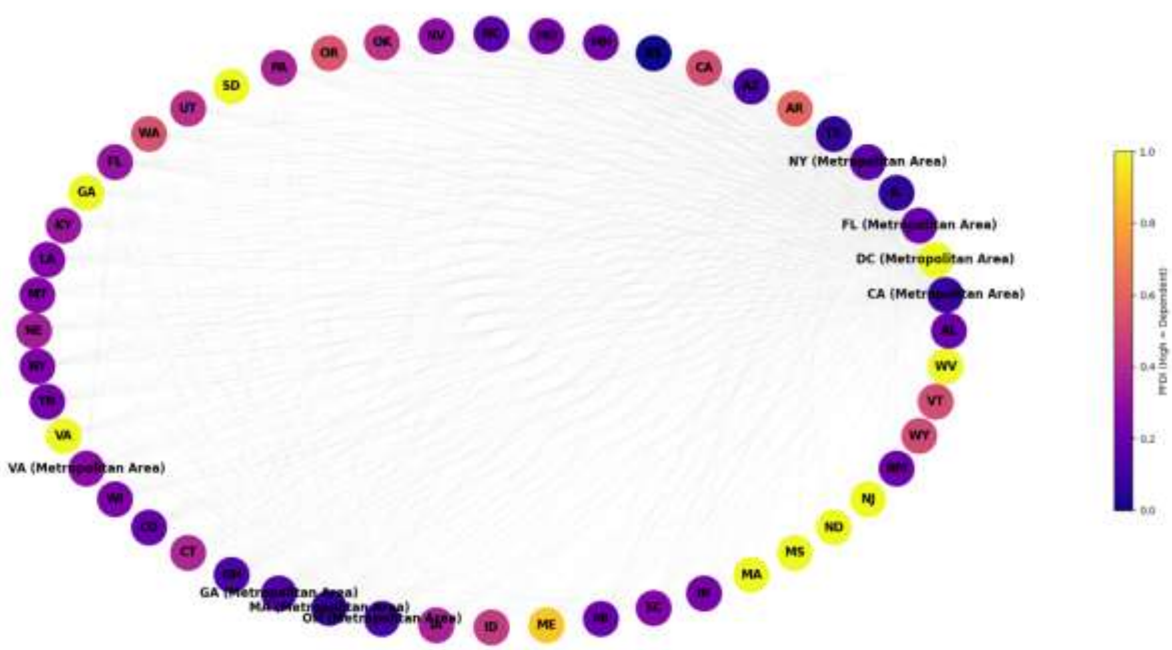
- These states have **some diversification:** passenger flows are distributed across multiple connections.
- Diversity values between **0.38–0.48** indicate **moderate local resilience**

Interpretation:

In 2020, the PFDI results indicate that while several small and metropolitan states are fully dependent on a single route, major hubs like California, Massachusetts, and Oregon maintain moderate diversification, highlighting structural resilience in parts of the U.S. airline network.

Early Recovery (Year 2021)

	State	PFDI	Diversity
2	DC (Metropolitan Area)	1.000000	0.000000
18	SD	1.000000	0.000000
29	VA	1.000000	0.000000
22	GA	1.000000	0.000000
44	MA	1.000000	0.000000
47	NJ	1.000000	0.000000
45	MS	1.000000	0.000000
51	WV	1.000000	0.000000
46	ND	1.000000	0.000000
40	ME	0.881136	0.118864



States with PFDI $= 1.0$

- DC, SD, VA, GA, MA, NJ, MS, WV, ND
- These states show **complete dependence on a single outgoing route**, with Diversity $= 0$.
- This indicates they are **highly fragile hubs**, where any disruption in their main route could significantly impact passenger flows.

State with moderate PFDI

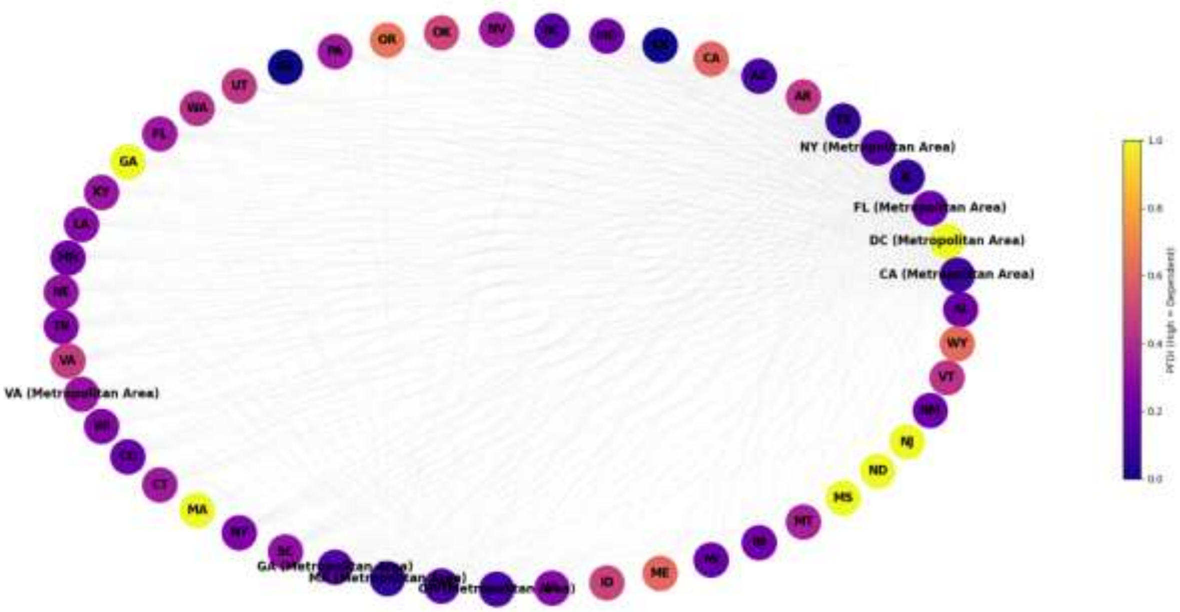
- ME (0.881) has a slightly lower PFDI, with Diversity $= 0.119$
- This indicates **partial diversification**, meaning ME's outgoing passengers are somewhat distributed across other connections.

Interpretation:

In 2021, the PFDI results indicate that several states, including DC, VA, and NJ, remain completely dependent on a single route, maintaining extreme fragility, while Maine shows partial diversification, highlighting initial improvements in resilience for some peripheral states.

Post−COVID−19 (Year 2022)

	State	PFDI	Diversity
2	DC (Metropolitan Area)	1.000000	0.000000
32	MA	1.000000	0.000000
21	GA	1.000000	0.000000
45	MS	1.000000	0.000000
47	NJ	1.000000	0.000000
46	ND	1.000000	0.000000
15	OR	0.648987	0.351013
50	WY	0.620017	0.379983
41	ME	0.612368	0.387632
9	CA	0.601796	0.398204



States with PFDI = 1.0

- DC, MA, GA, MS, NJ, ND
- These states are **fully dependent on a single outgoing route**, with Diversity = 0.
- This indicates **extreme fragility**, meaning any disruption in their main route could severely impact passenger flows.

States with moderate PFDI (0.60–0.65)

- OR (0.649), WY (0.620), ME (0.612), CA (0.602)

- These states have **partial diversification**, with passenger flows distributed across several routes.
- Diversity values between 0.35–0.40 indicate **moderate resilience**, able to better handle disruptions compared to fully dependent states.

Interpretation:

In 2022, the PFDI results reveal that several states, including DC, MA, and NJ, remain fully dependent on a single route, maintaining high fragility, while states like California, Oregon, Maine, and Wyoming show moderate diversification indicating improved resilience in key hubs.

Overall Interpretation:

From 2019 to 2022, the U.S. airline network shows a dual structure: metropolitan hubs like DC, NJ, and MA remain fully dependent on a single route (high PFDI, low Diversity), making them fragile, while major hubs like California, Oregon, and Maine maintain moderate diversification, providing resilience. Over time, slight increases in Diversity indicate gradual improvement in network robustness.

Conclusion:

This project demonstrates that the U.S. domestic airline network is structurally resilient yet highly centralized, relying heavily on a small set of dominant hub states to sustain national connectivity. Across the 2019–2022 period, Social Network Analysis consistently identifies California, Texas, Florida, New York, and Illinois as the core anchors of passenger movement, functioning simultaneously as major hubs, bottlenecks, and influential spreaders of connectivity and disruption. The persistence of these states across all centrality measures confirms the stability of the U.S. hub-and-spoke system, even under extreme external shocks.

The COVID-19 pandemic introduced a sharp but temporary contraction in network activity, intensifying centralization and increasing dependency on fewer intermediary states. During this period, regional airline structures gained importance, as states with high clustering coefficients maintained short-range connectivity and contributed to network stability when long-distance routes collapsed. As recovery progressed in 2021 and 2022, the network largely reverted to its pre-pandemic hierarchy, though with increased consolidation around the largest hubs—suggesting a “winner-take-all” dynamic in post-crisis air travel.

Passenger Flow Dependency and Diversity analysis further reveals a dual structure within the network: while major hubs exhibit moderate diversification and resilience, several metropolitan and smaller states remain fully dependent on a single route, making them structurally fragile. This imbalance highlights an important vulnerability in the U.S. airline system, where efficiency is achieved at the cost of robustness for peripheral states.

Overall, the findings underscore that the resilience of the U.S. airline network depends not only on powerful national hubs but also on the strength of regional connectivity and route diversification. From a policy and planning perspective, strengthening alternative routes and reducing extreme dependency could enhance systemic robustness against future disruptions. Social Network Analysis proves to be a valuable framework for uncovering these structural insights, offering a deeper understanding of how mobility systems absorb shocks, adapt, and recover over time.

List of reference:

Dataset

- US Airline Flight Routes and Fares 1993–2024 Dataset
- This dataset includes historical U.S. airline flight routes, allowing the construction of yearly networks for 2019, 2020, 2021, and 2022.
- Link US Airline Flight Routes and Fares 1993-2024

Tools

- **Gephi:** Used for network visualization and calculating Social Network Analysis (SNA) metrics.
- **NetworkX:** A Python library used alongside Gephi to compute metrics and perform structural analysis.

Academic Frameworks & Theories

- **Social Network Analysis (SNA):** The primary methodology used to model states as nodes and weighted passenger flows as directed edges.
- **Hub-and-Spoke Model:** The theoretical foundation for understanding how major states act as central hubs for passenger distribution.
- **Regional Development Theory:** Used to link air connectivity with localized economic activity and tourism.
- **Transportation Resilience Theory:** Applied to evaluate how the network absorbs shocks like COVID-19 and its subsequent recovery.

Key Metrics Used

- **Centrality Metrics:** Degree centrality, weighted in-degree (passengers landing), weighted out-degree (passengers departing), weighted betweenness centrality (bottlenecks), and eigenvector centrality (influence).
- **Closeness Centrality:** Utilized to identify states most likely to spread disruptions or delays widely across the network.
- **Clustering Coefficient:** Applied to measure short-range regional connectivity and local network stability.
- **Degree Assortativity:** Used to determine whether major hubs connect primarily with other hubs or smaller regional states.
- **Passenger Flow Dependency Index (PFDI) and Diversity Index (PFDI-D):** Custom indices developed to measure route concentration and the structural fragility or resilience of individual states.

9) Task contribution:

	Data Collection	Data Cleaning	Gephi Analysis	Results	Report	Presentation
Salma Khaled	✓	✓	✓	✓	✓	
Reham Mohamed	✓		✓	✓	✓	
Mona Ahmed	✓	✓	✓	✓		
Menna Mostafa		✓	✓	✓	✓	
Nour Osama			✓	✓	✓	✓